Random Variables and Linear Mixed Effects Models

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Dartmouth College. February 2026

Variation in Linear Regressions

When we do a linear regression, we are trying to explain **variation**.

We have factors we understand and we can test for
These are the independent variables.

We have factors we do not know and cannot test for.
Usually we don't have any information about these, but sometimes we do!

E.g. `Subject`
`Item`

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E.g. `Subject`
`Item`

We call these factors **random effects**.
(Because they are not predictable, like the independent variables we are trying to control)

Random Effects Example 1: RT

In English, reaction time can be predicted by:

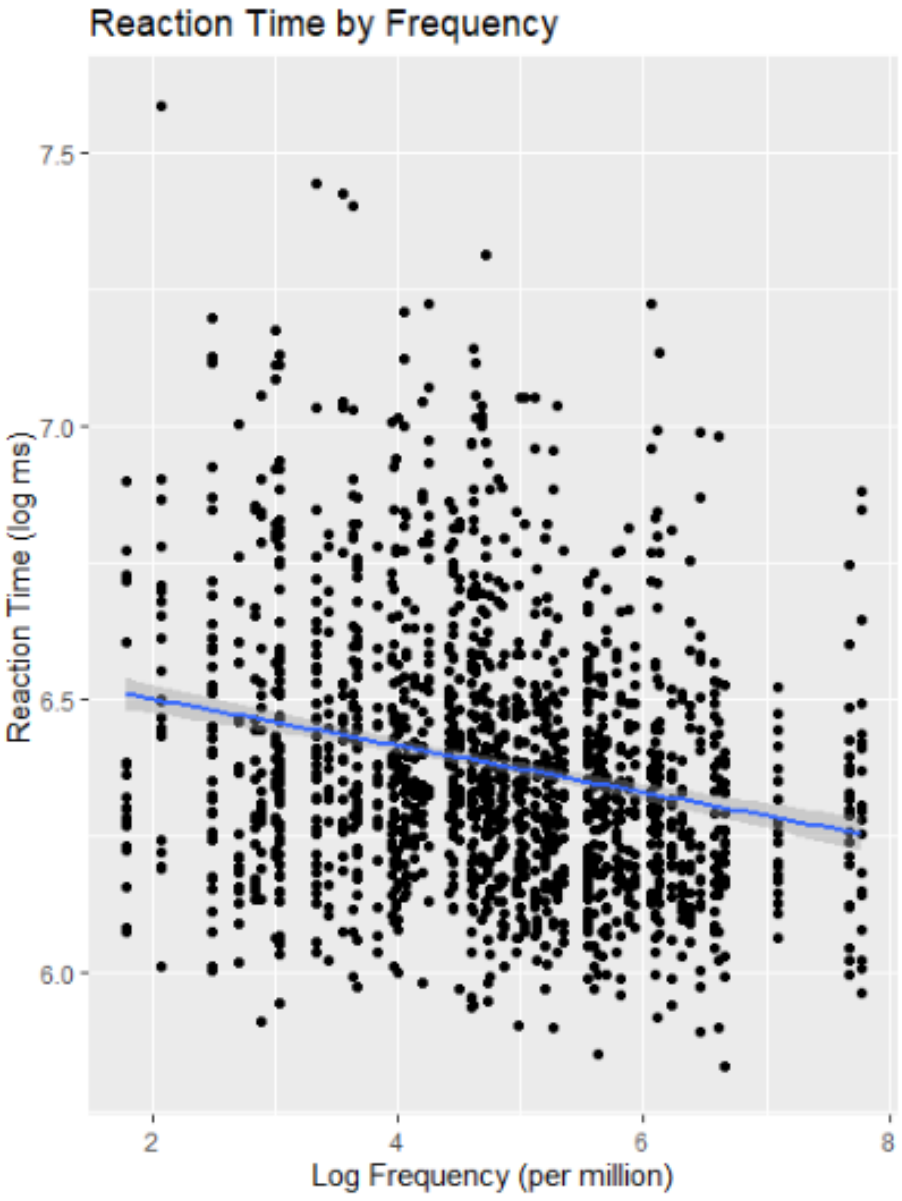
`Length`	Length of the word, in letters
`Frequency`	Frequency (in words per million)

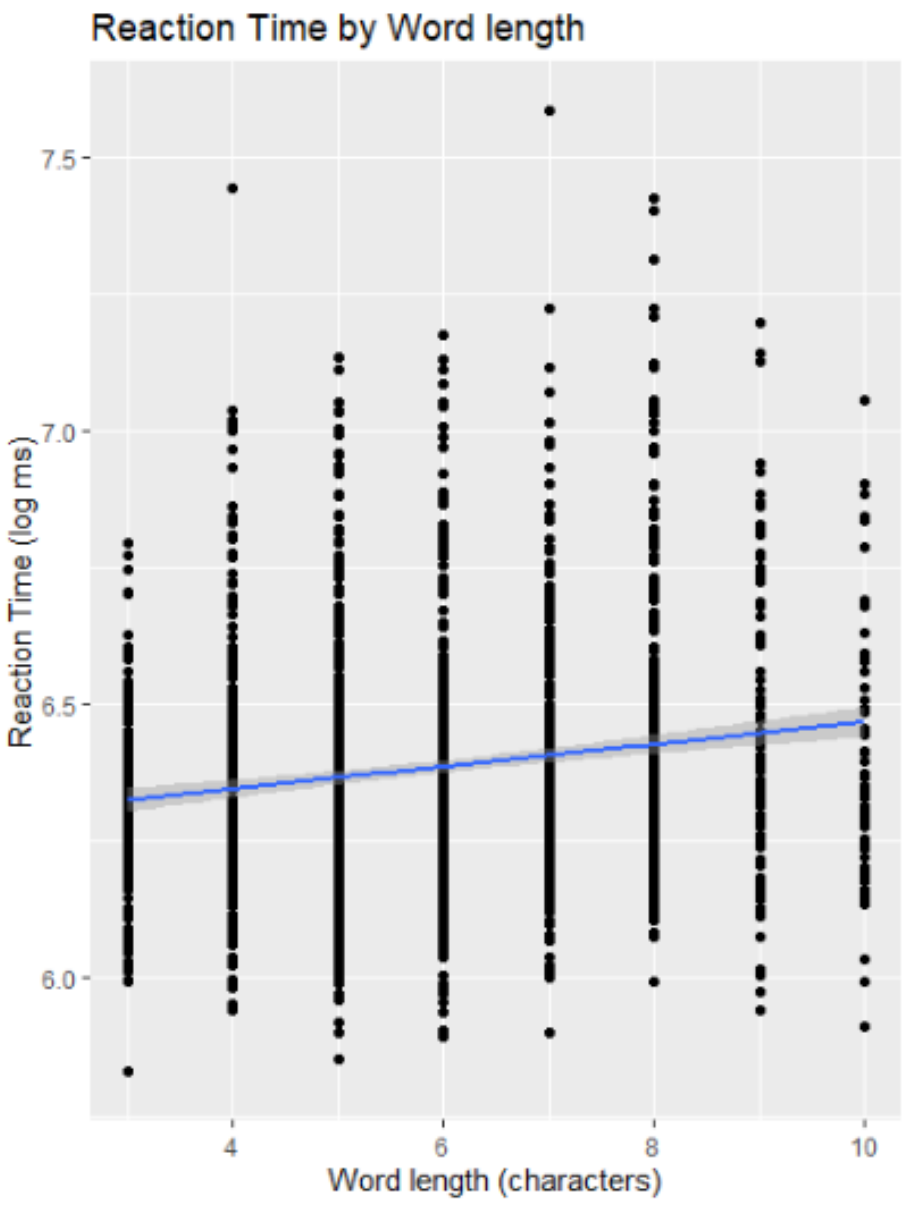
In addition to this, the dataset lexdec.csv also contains:

`Subject`	Which person gave which answers
`Word`	Which word elicited the reaction time

What is the relationship between these four variables and `RT`?

Random Effects Example 1: RT





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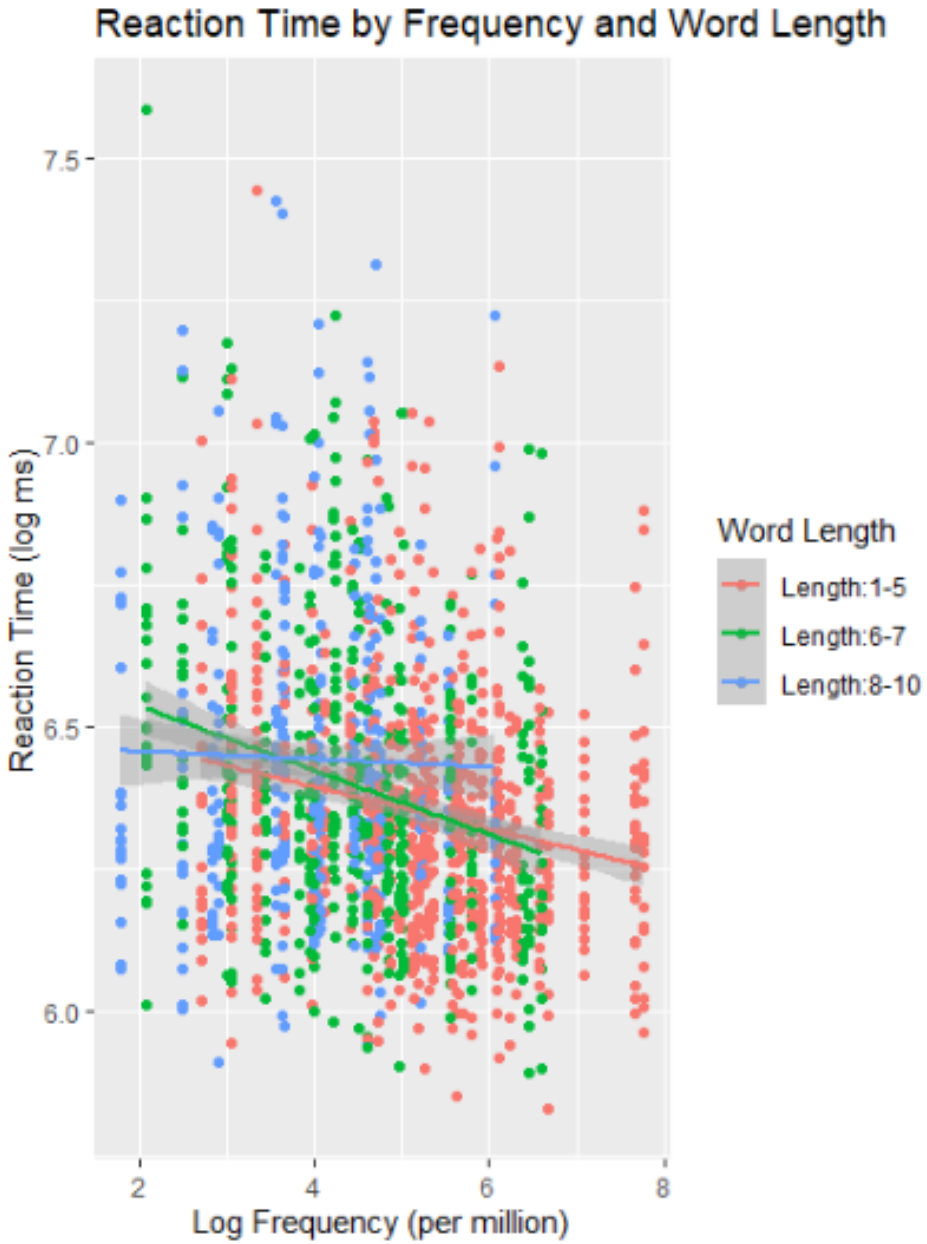

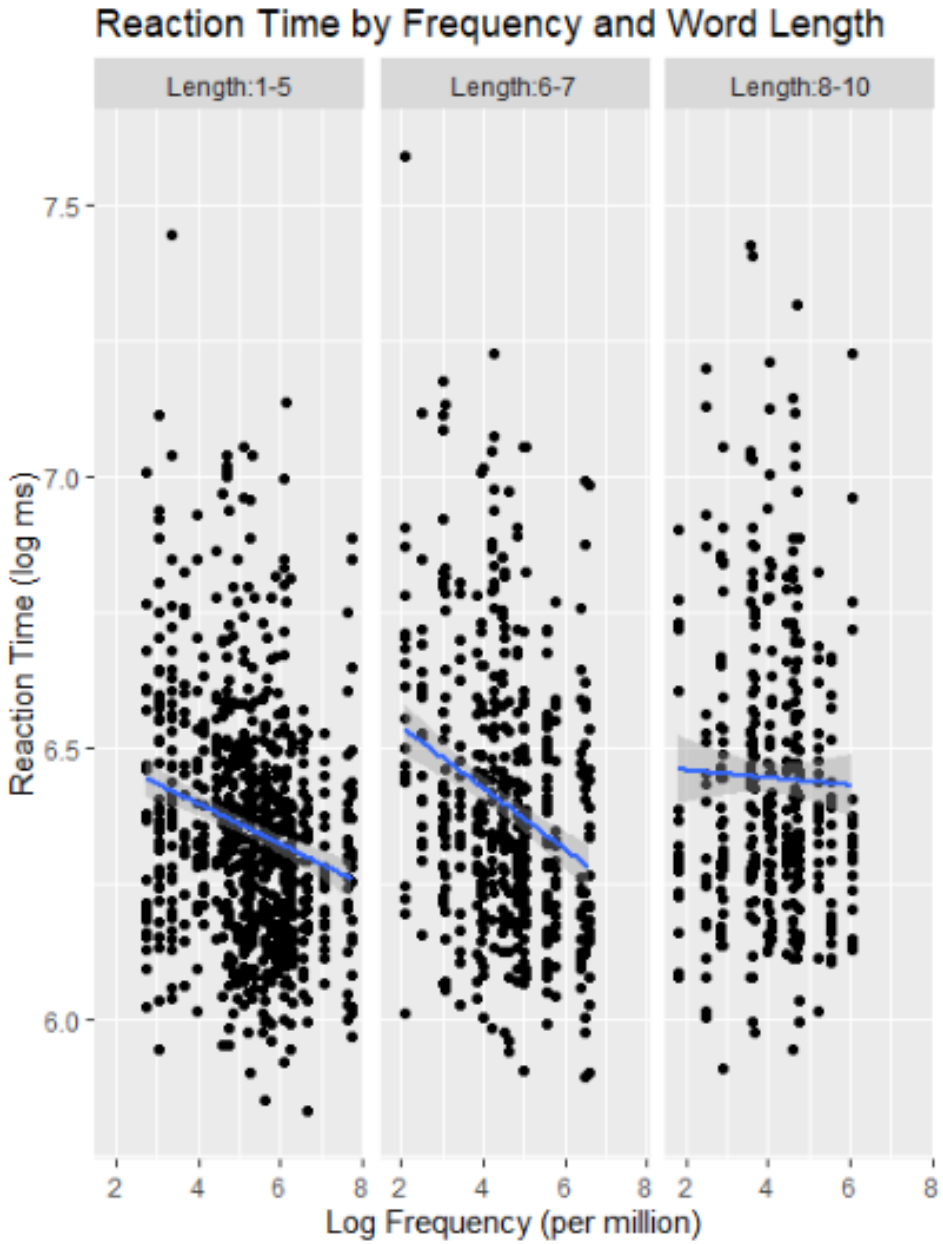

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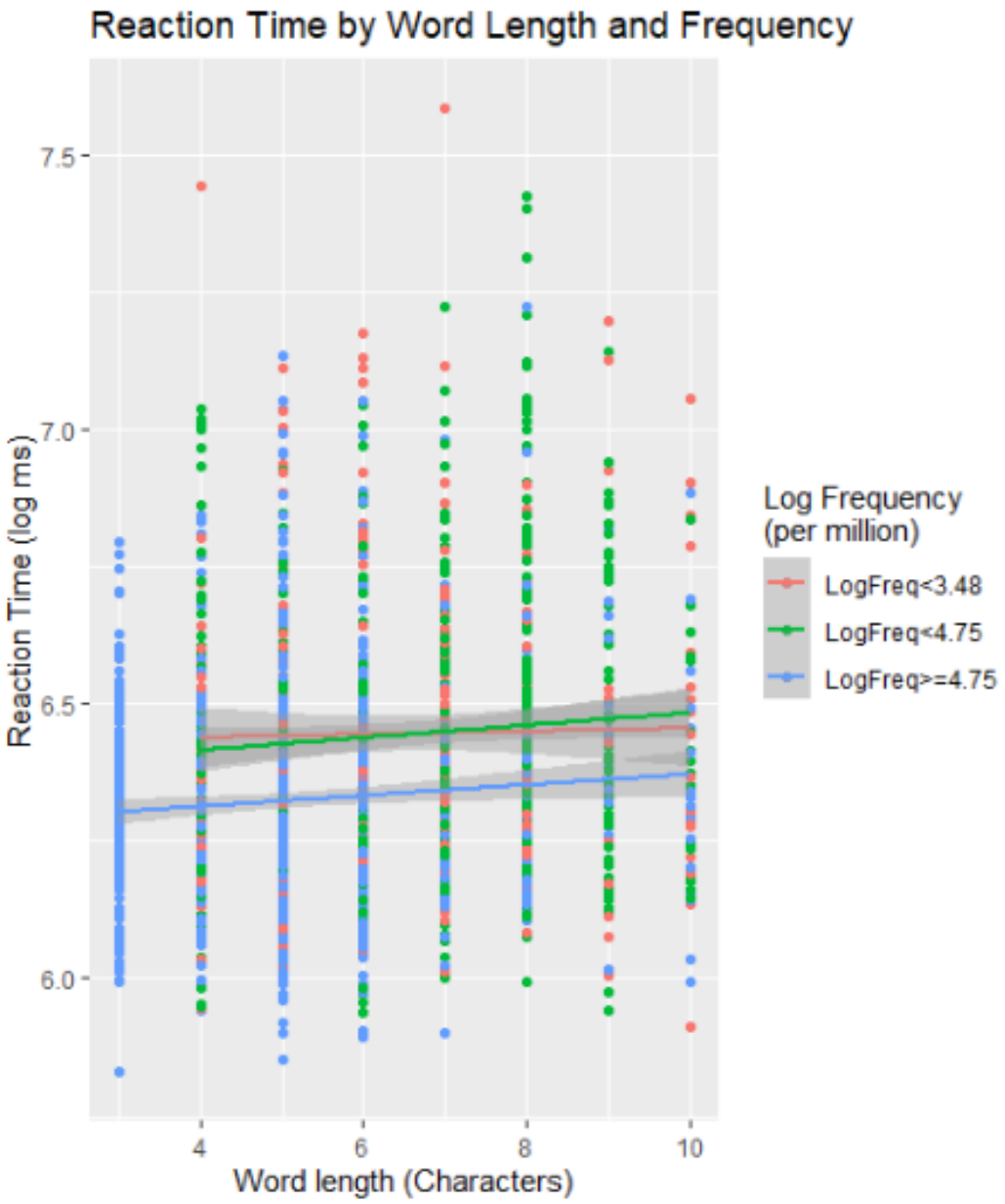

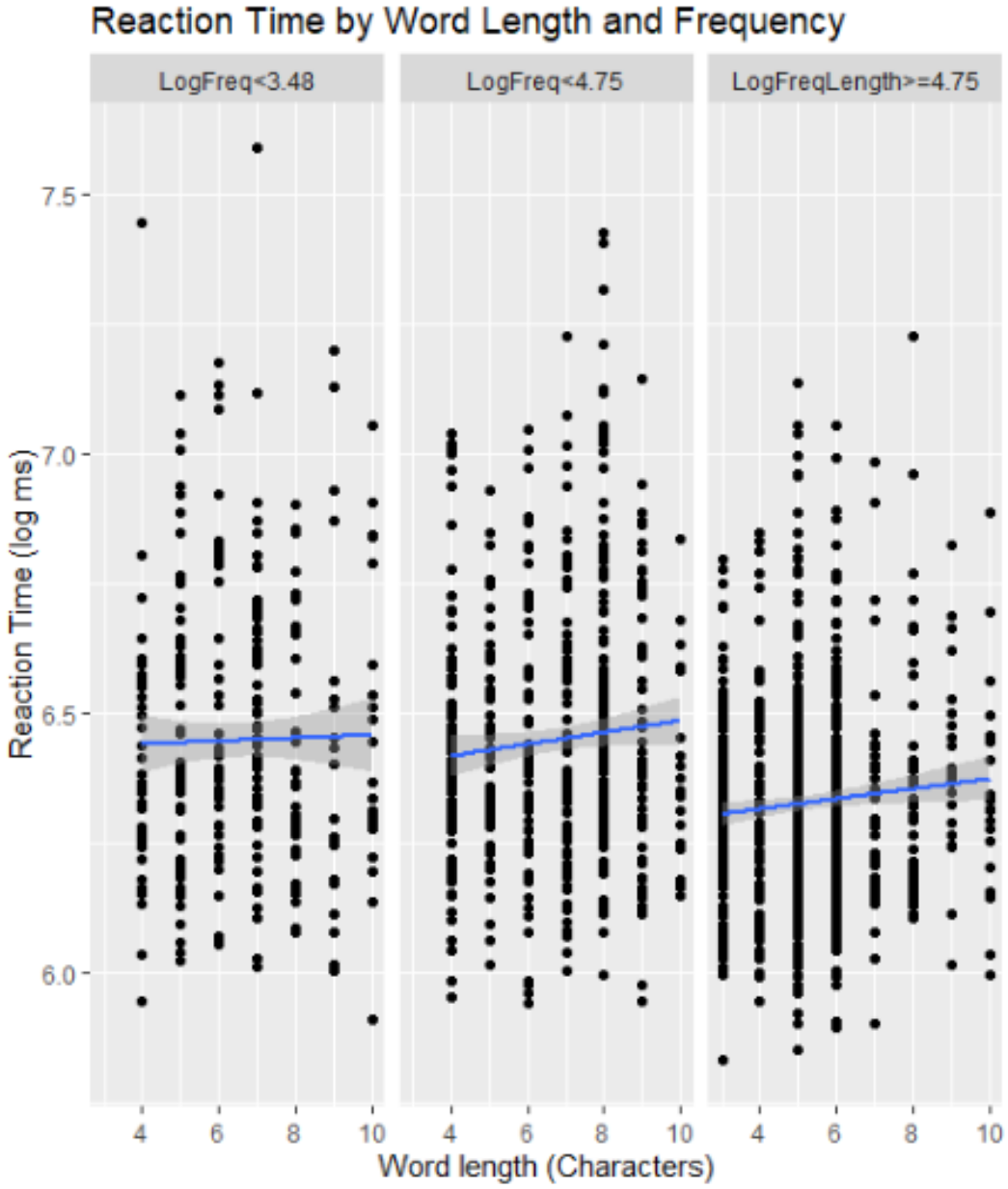

Random Effects Example 1: RT

```
> mLex1 <- lm(RT ~ Length + Frequency, lex)
> summary(mLex1)
Call:
lm(formula = RT ~ Length + Frequency, data = lex)
Residuals:
     Min       1Q   Median       3Q      Max 
-0.58141 -0.16153 -0.03286  0.12343  1.09312 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  6.505756   0.037783 172.189  < 2e-16 ***
Length       0.009348   0.003438   2.719  0.00662 ** 
Frequency   -0.037028   0.005008  -7.393 2.26e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2349 on 1656 degrees of freedom
Multiple R-squared:  0.05544,	Adjusted R-squared:  0.0543 
F-statistic:  48.6 on 2 and 1656 DF,  p-value: < 2.2e-16
```

Random Effects Example 1: RT

```
> mLex2 <- lmer(RT ~ Length + Frequency + (1+Frequency|Word) +
(1+Frequency|Subject), lex)
```

(1) New instruction: `lmer` (Linear mixed-effects)
(2) Random effects: `Word`, `Subject`
(3) Random intercept: 1
(4) Random slope: `Frequency`

Random Effects Example 1: RT

```
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Frequency + (1+Frequency|Word)
+ (1+Frequency|Subject), lex)
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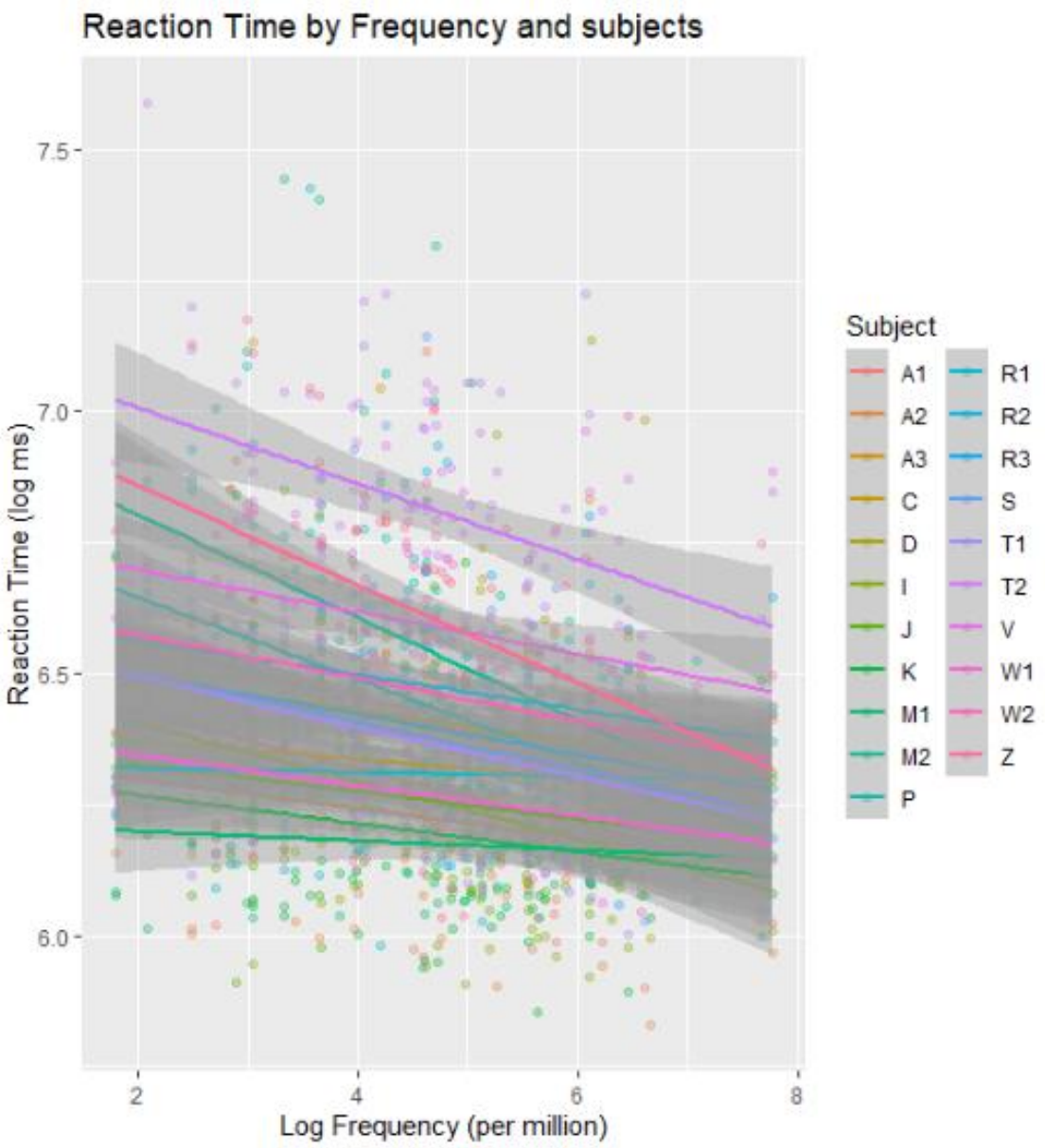

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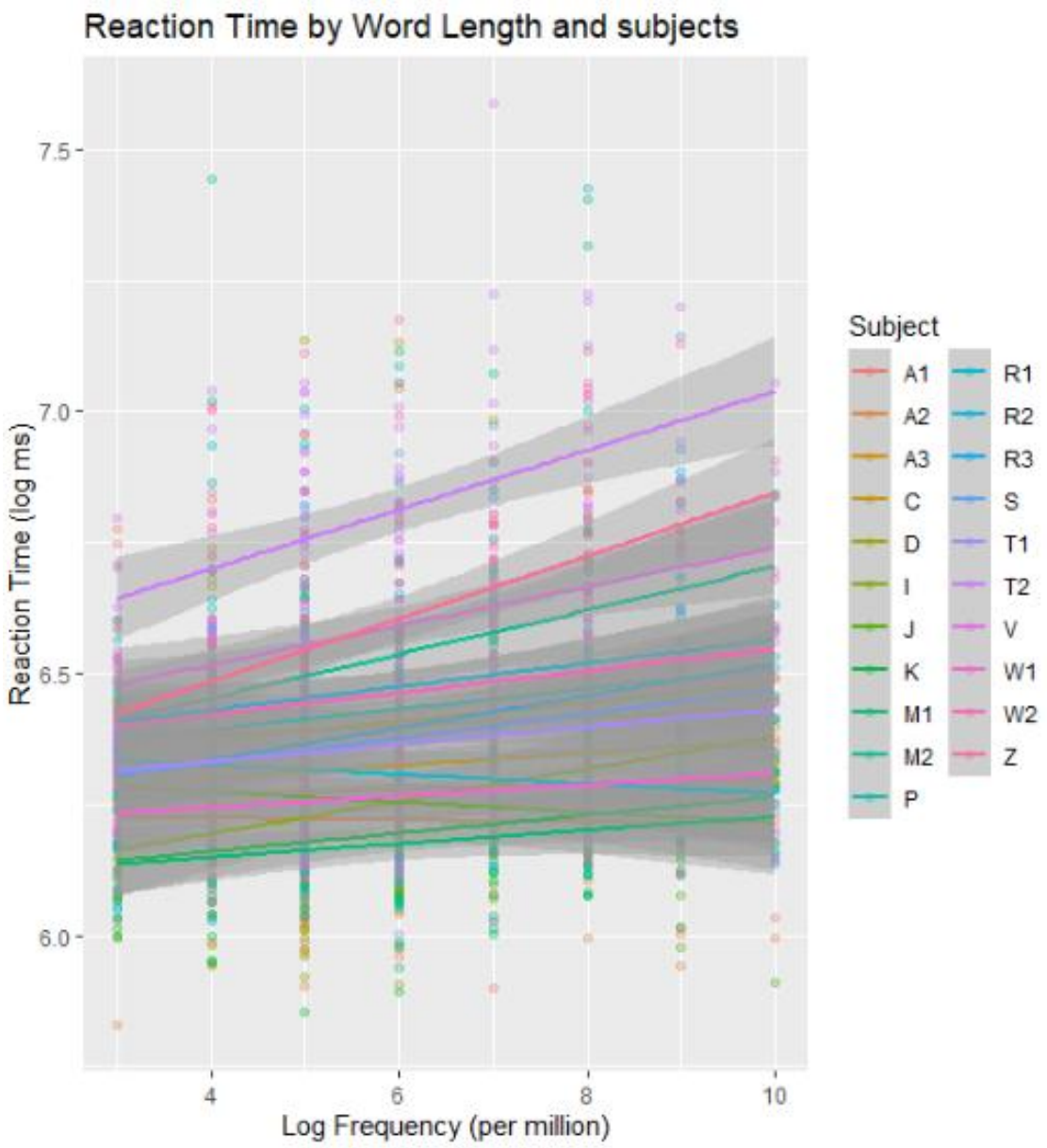

Random Effects Example 1: RT

```
> mLex2 <- lmer(RT ~ Length + Frequency + (1+Frequency|Word) +
(1+Frequency|Subject), lex)

Random effects:
 Groups   Name        Variance  Std.Dev. Corr
 Word     (Intercept) 0.0356060 0.18870
          Frequency   0.0009332 0.03055  -1.00
 Subject  (Intercept) 0.0563874 0.23746
          Frequency   0.0004094 0.02023  -0.92
 Residual             0.0291296 0.17067
Number of obs: 1659, groups:  Word, 79; Subject, 21
```

Random Effects Example 1: RT

```
> mLex2 <- lmer(RT ~ Length + Frequency + (1+Frequency|Word) +
(1+Frequency|Subject), lex)

Fixed effects:
             Estimate Std. Error         df t value Pr(>|t|)
(Intercept)  6.523945   0.070433  50.440873  92.626  < 2e-16 ***
Length       0.010209   0.003750 145.469172   2.723  0.00727 **
Frequency   -0.042342   0.007908  47.240314  -5.354 2.48e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
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A linear mixed-effects model was used, with reaction time as the dependent variable, length (in letters) and frequency (in words per million) as independent factors, and item and subject as random effects (with random intercepts and a random slope for Frequency). Longer words have slower RT ($\beta=0.01 \pm 0.004$, $t(145)=2.7$, $p<0.01$), while more frequent words have faster RT ($\beta=-0.04 \pm 0.008$, $t(47)=-5.4$, $p<0.0001$).

Random Effects Example 1: RT

```
> mLex2 <- lmer(RT ~ Length + Frequency + (1+Frequency|Word) +
(1+Frequency|Subject), lex)
```

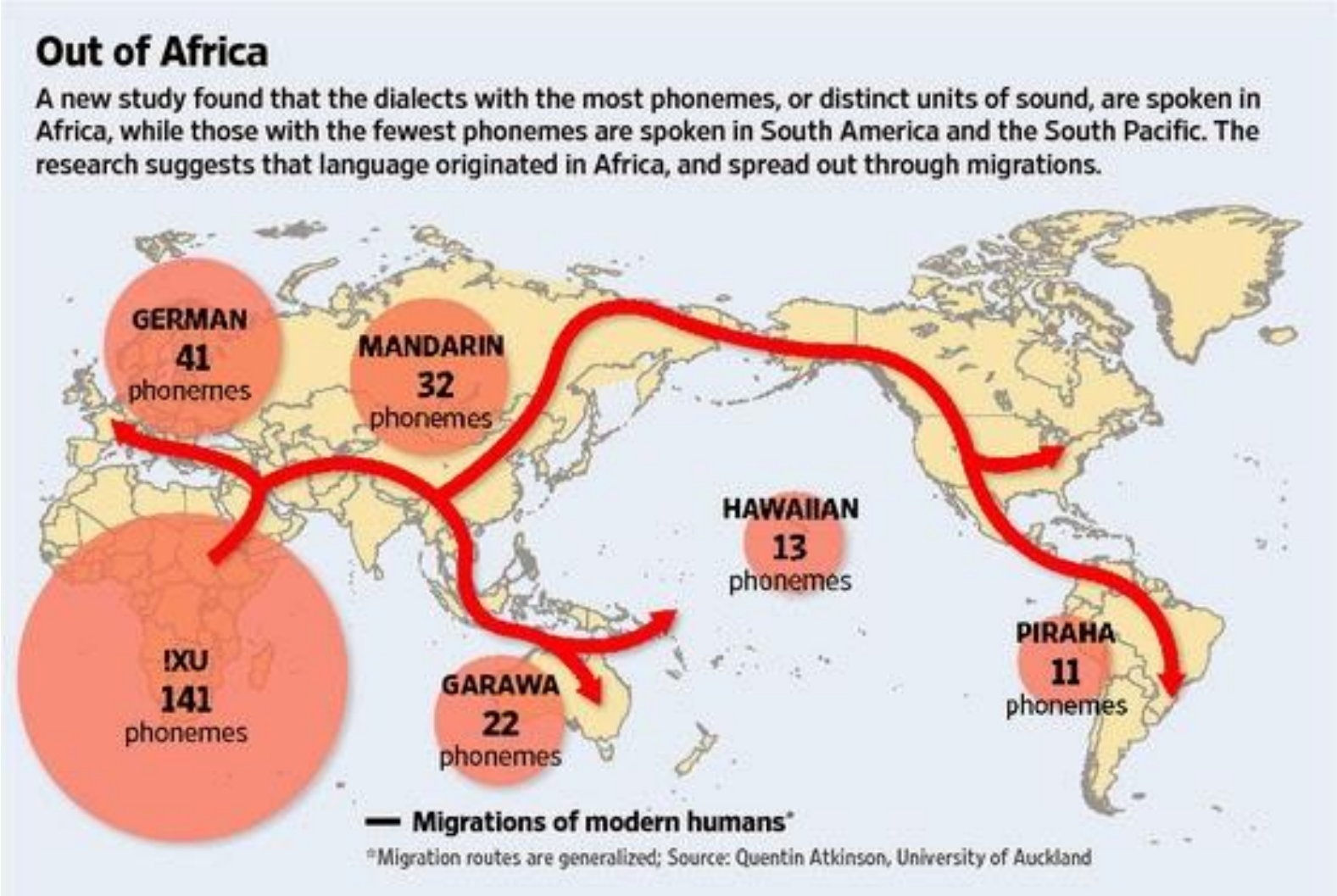
Internally, the equation is:

$$RT_{(i)} = \alpha + \beta_1 Length + \beta_2 Frequency + \alpha_{subject(i)} + \beta_{subject(i)} Frequency + \alpha_{word(i)} + \beta_{word(i)} Frequency$$

It adds an additional term when making the prediction for each specific word. It adds a little "oomph" to the intercept and to the slope.

Variation in Linear Regressions

Another reason why we use random effects is that sometimes global trends are deceiving!



Atkinson (2011): Phonemic Diversity supports a serial founder effect model of language expansion from Africa. Science 332(6027).

New York Times: Phonetic Clues Hint Language Is Africa-Born

Variation in Linear Regressions

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REPORT

Phonemic Diversity Supports a Serial Founder Effect Model of Language Expansion from Africa

QUENTIN D. ATKINSON

SCIENCE • 15 Apr 2011 • Vol 332, Issue 6027 • pp. 346-349 • DOI: 10.1126/science.1199295

Abstract

Human genetic and phenotypic diversity declines with distance from Africa, as predicted by a serial founder effect in which successive population bottlenecks during range expansion progressively reduce diversity, underpinning support for an African origin of modern humans. Recent work suggests that a similar founder effect may operate on human culture and language. Here I show that the number of phonemes used in a global sample of 504 languages is also clinal and fits a serial founder–effect model of expansion from an inferred origin in Africa. This result, which is not explained by more recent demographic history, local language diver-

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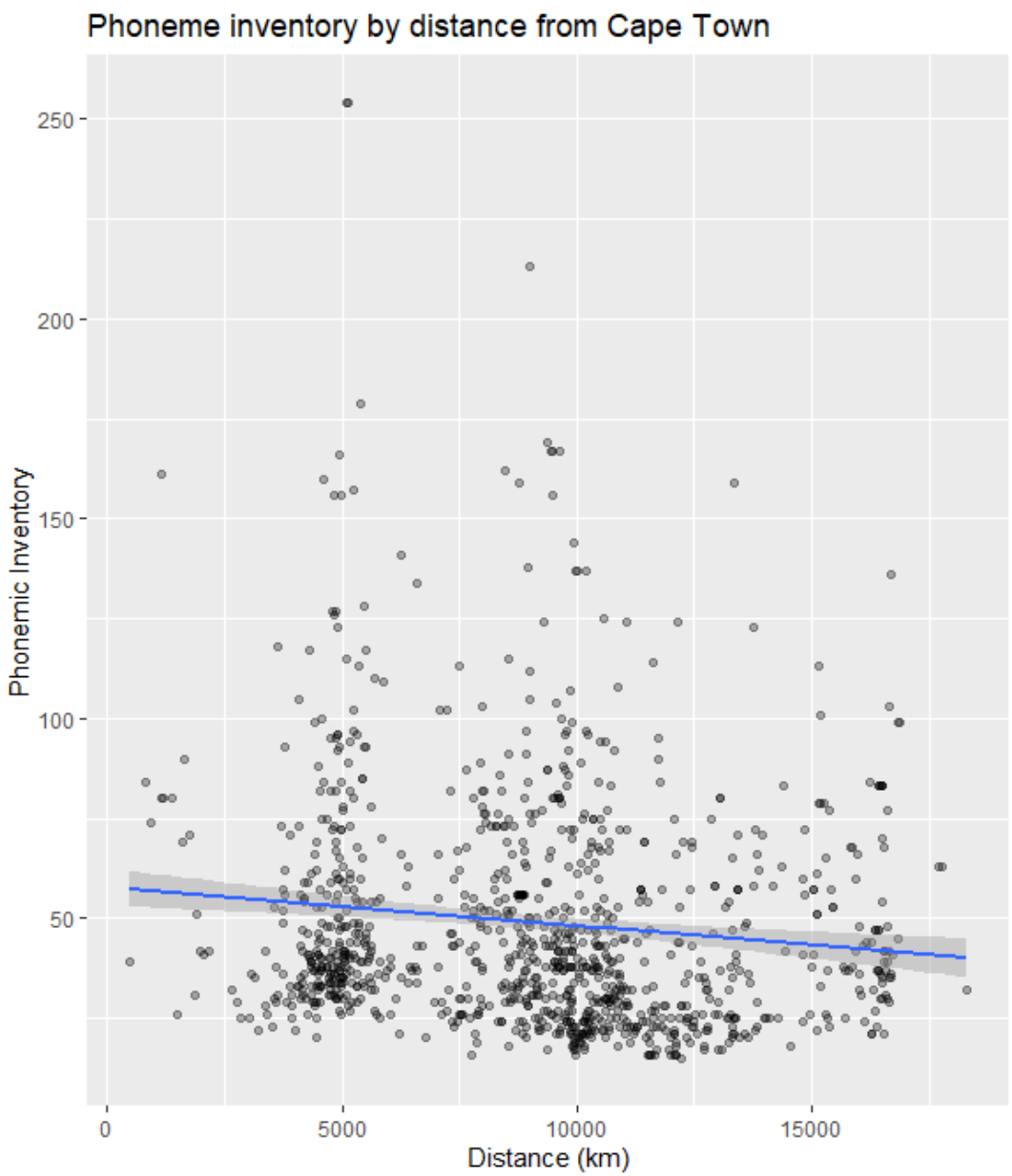

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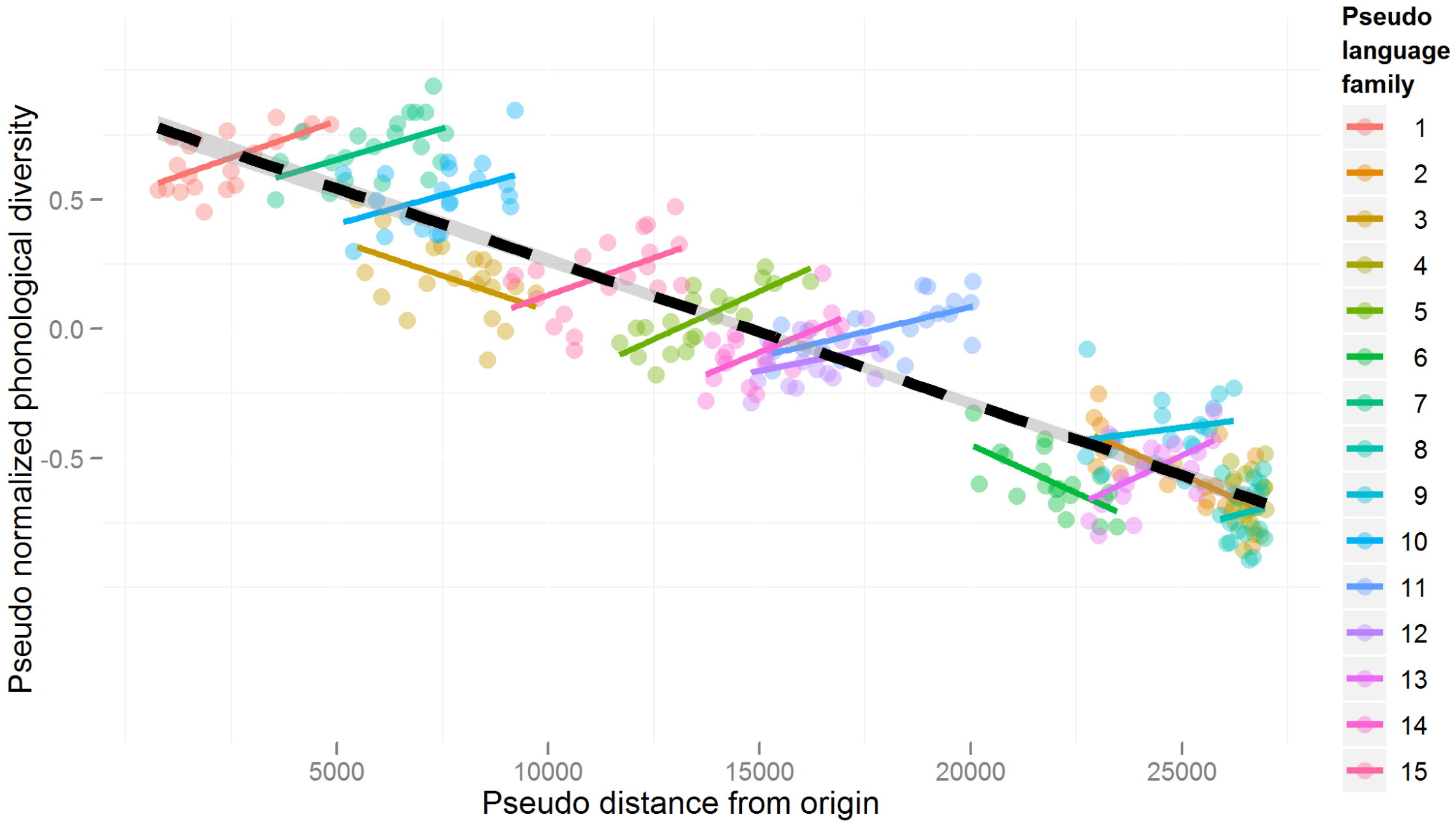

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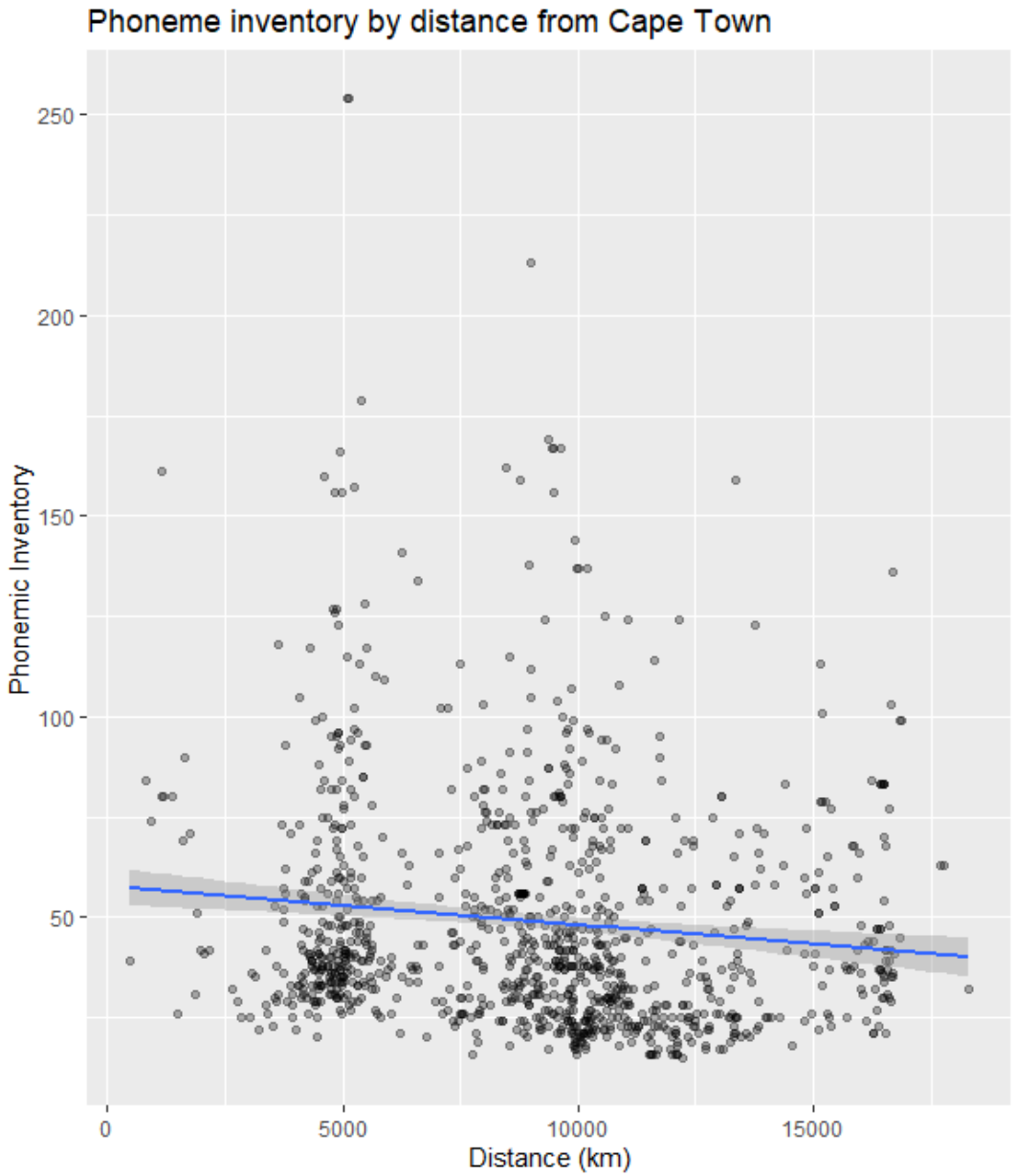

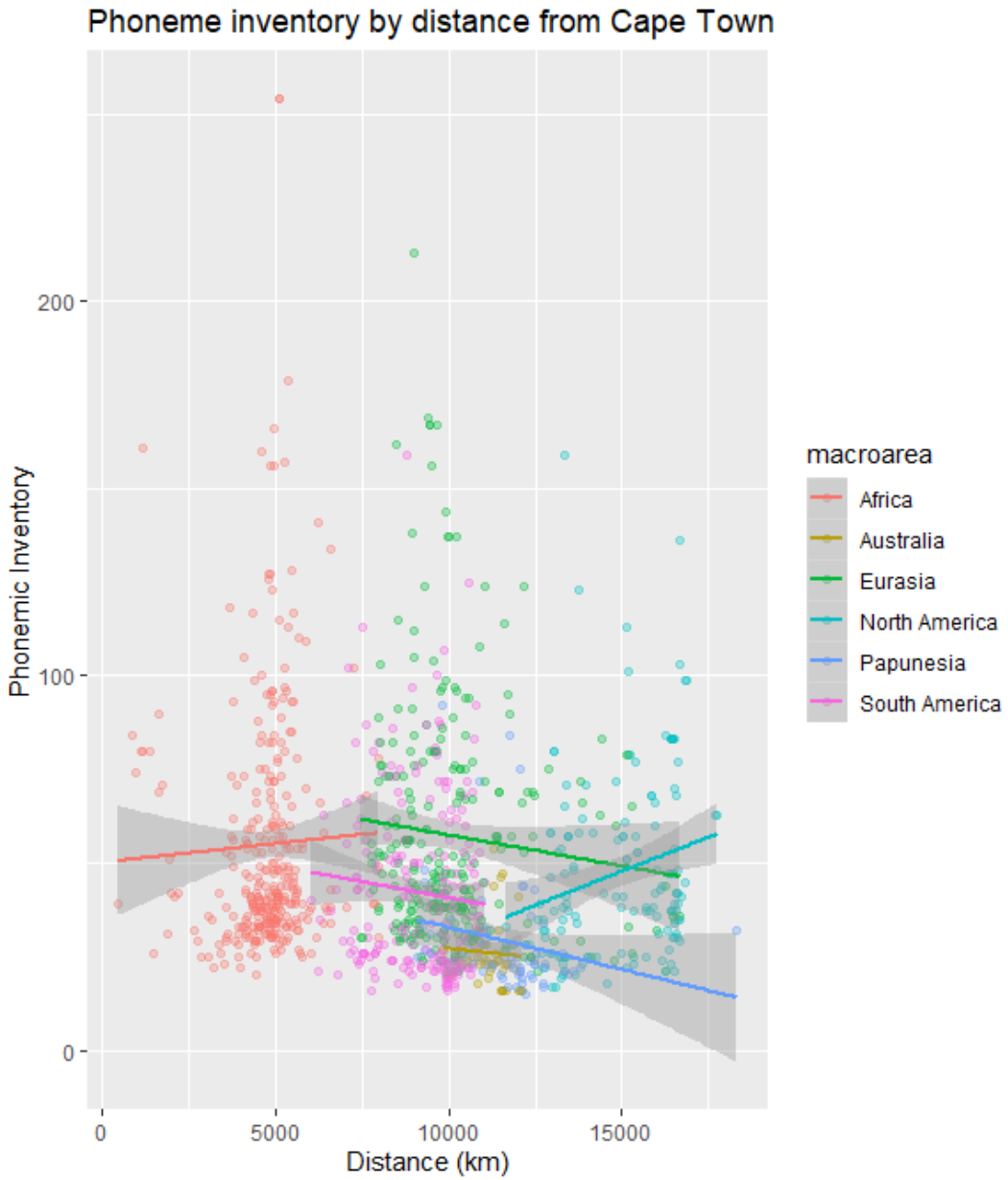

Variation in Linear Regressions

Random effects can add explanatory power to our regression.

We can now explain variance from:

- The variables we are actually studying
- Known environmental factors (e.g. word, subject, language family, geographical area).

Confounding factors and bad correlations

Storks bring babies

◆ TESTING THE STORK-BIRTH RELATIONSHIP ◆

Country	Area (km^2)	Storks (pairs)	Humans (10^6)	Birth rate (10^3/yr)
Albania	28,750	100	3.2	83
Austria	83,860	300	7.6	87
Belgium	30,520	1	9.9	118
Bulgaria	111,000	5000	9.0	117
Denmark	43,100	9	5.1	59
France	544,000	140	56	774
Germany	357,000	3300	78	901
Greece	132,000	2500	10	106
Holland	41,900	4	15	188
Hungary	93,000	5000	11	124
Italy	301,280	5	57	551
Poland	312,680	30,000	38	610
Portugal	92,390	1500	10	120
Romania	237,500	5000	23	367
Spain	504,750	8000	39	439
Switzerland	41,290	150	6.7	82
Turkey	779,450	25,000	56	1576

Table 1. Geographic, human and stork data for 17 European countries

H_0 (null hypothesis): There is no relationship between babies and storks.

`b=0` in `birthRate = a + b*storks`

H_1 (alternate hypothesis): There is some relationship between babies and storks.

`b≠0` in `birthRate = a + b*storks`

Matthews, R. (2000). Storks deliver babies (p= 0.008). Teaching Statistics, 22(2), 36-38.

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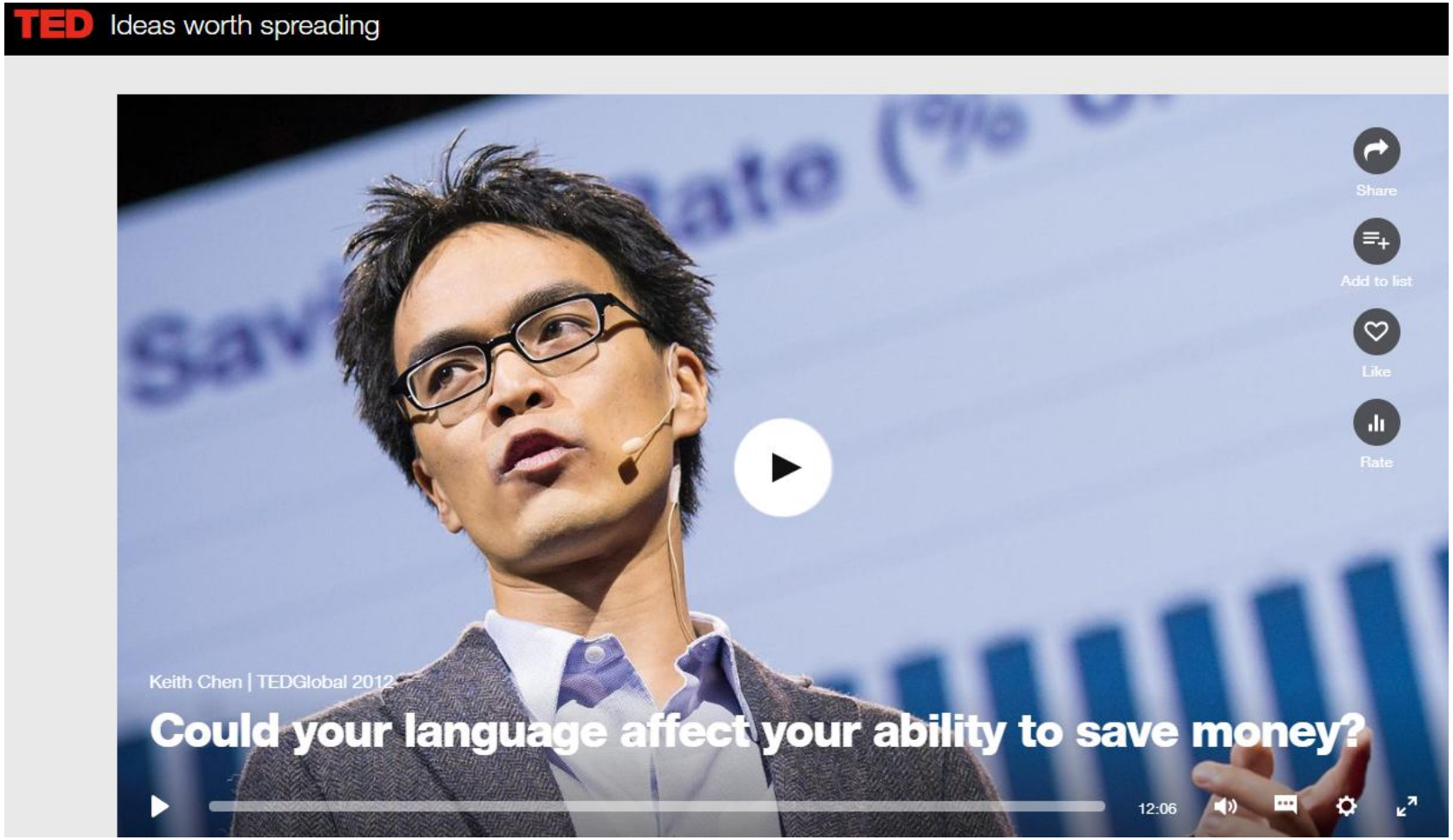
n = 17

t(15)= 3.06, p = 0.008

R^2 = 0.38

```
birthRate = 225 + 0.029(storks) + ε
```

False correlations

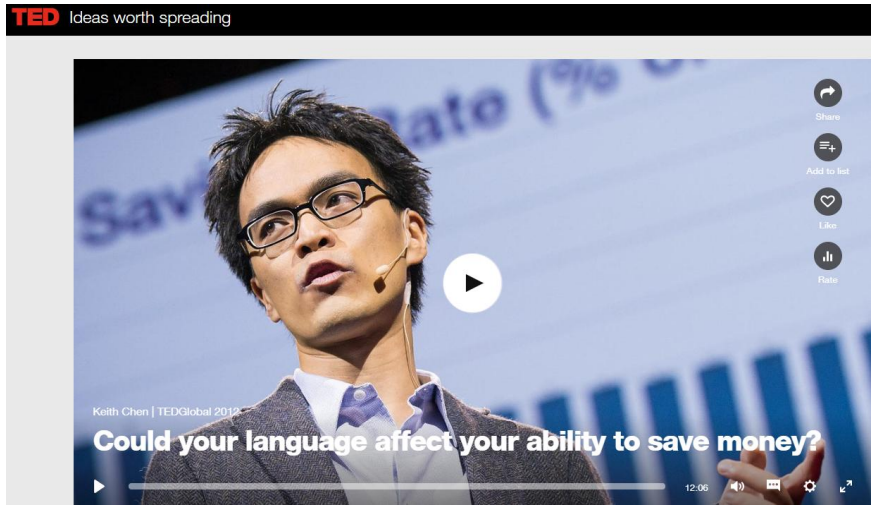


False correlations

Languages that make little or no distinction between the present and the future induce speakers to make more future-oriented choices.

Speakers of languages that do not distinguish between the present and the future save more money, retire with more wealth, smoke less, practice safer sex, and are less obese.

There's a connection between how you feel about the future and how your language forces you to talk about the future (Chen 2012) >>



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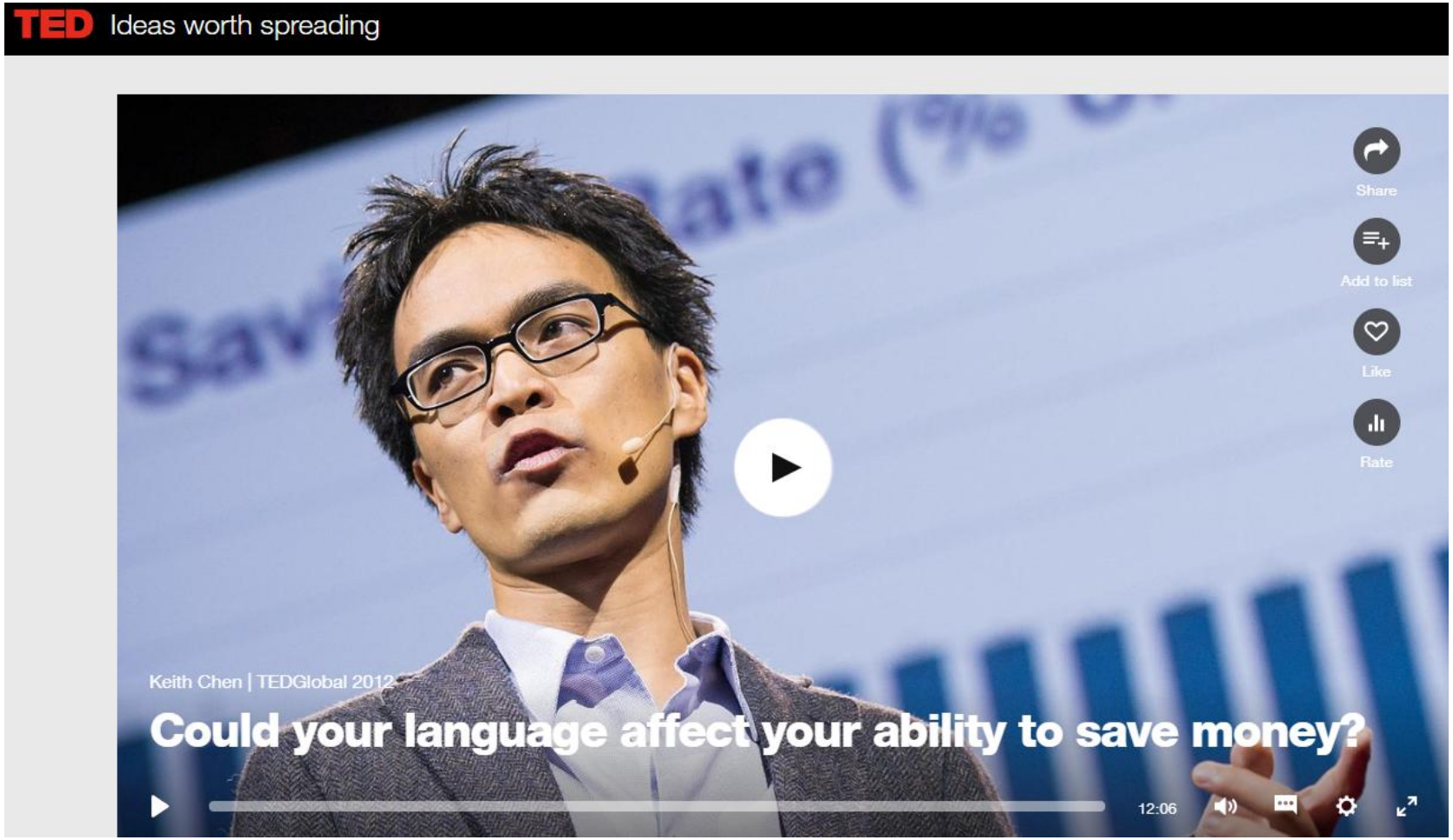
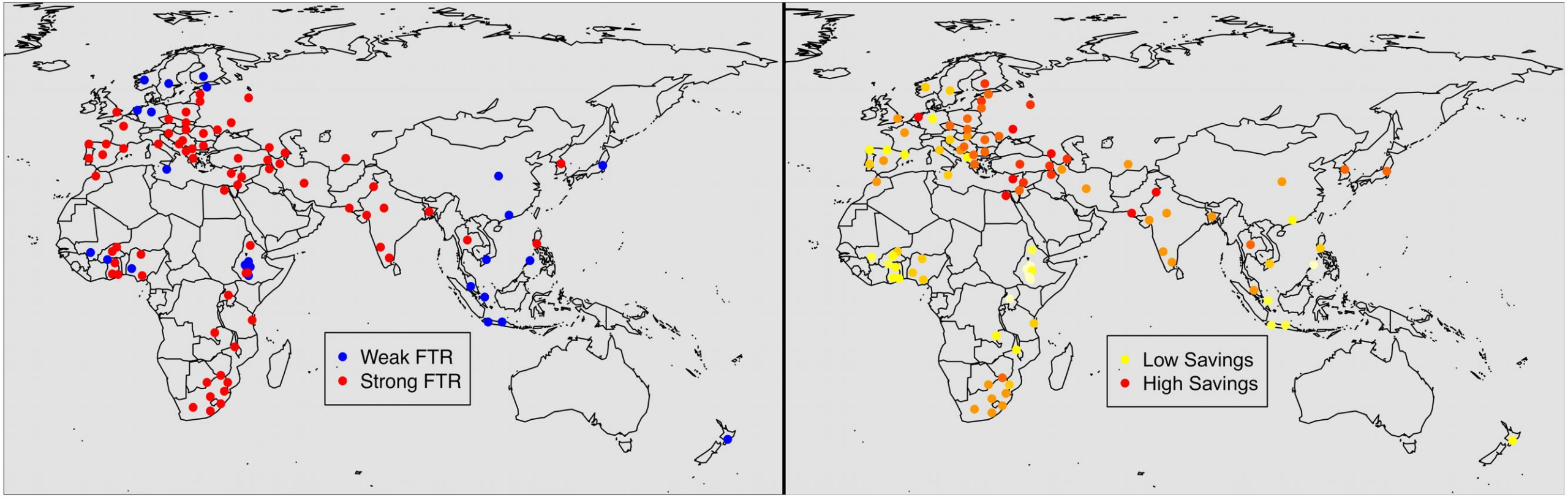


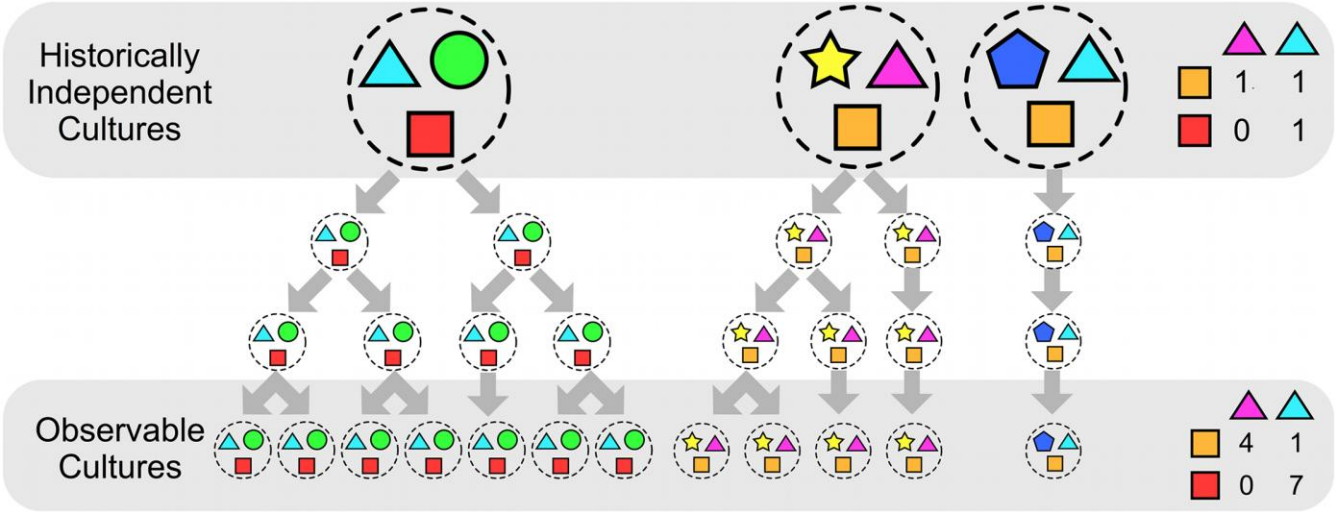
No. (Roberts et al. 2015 >>)

False correlations



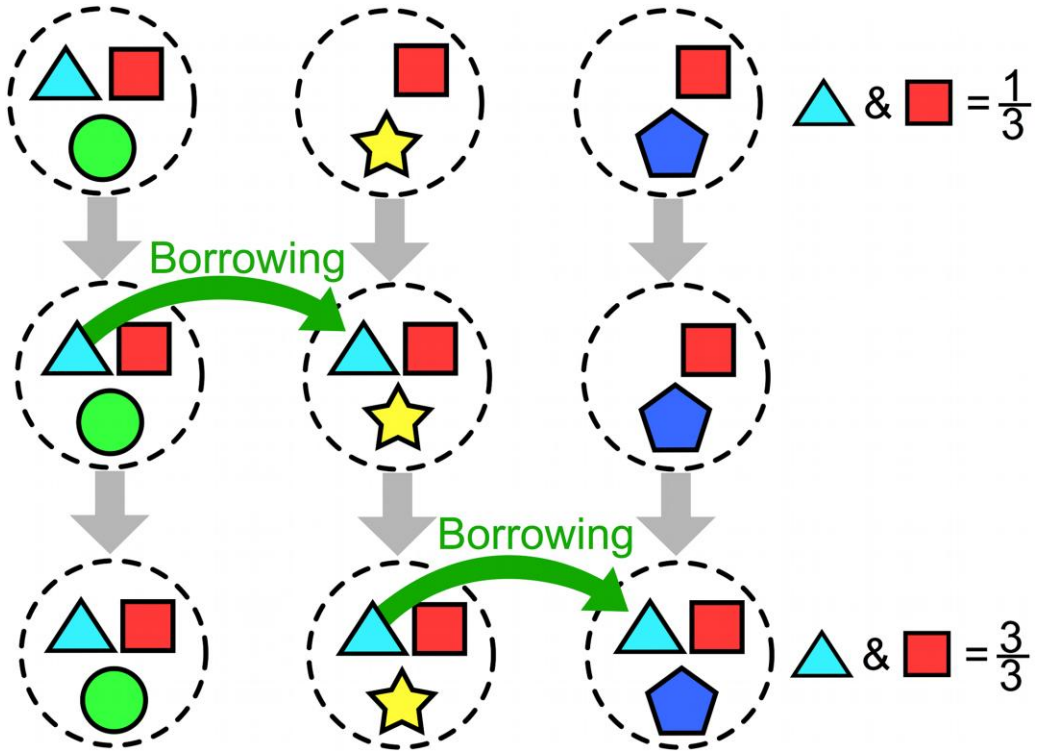
Chen 2012 >>

False correlations



Similarity due to relatedness

Similarity due to borrowing



False correlations

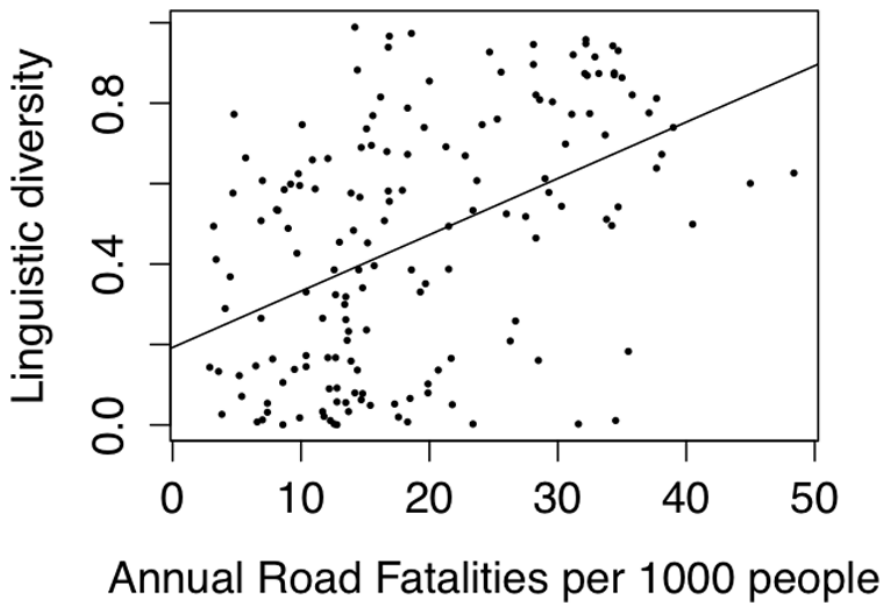
Test	Summary of test	Is data aggregated?	Control for language family	Control for geographic area	Control for country	Is the correlation robust?
Mixed effects model	Is the correlation robust when controlling for the random influence of language family, geographic area or country?	No	Yes	Yes	Yes	No
Regression on matched samples	Does FTR predict savings when comparing individuals that are matched on many levels, including language family?	No	Yes	No	Yes	Yes
Serendipity test	Is savings behaviour more strongly associated with FTR than other linguistic variables?	No	Yes	No	Yes	Yes
Independent samples	Do speakers of strong FTR languages have a lower average propensity to save in historically independent languages?	Yes	Yes	No	No	Yes
Partial Mantel test	Is the difference in saving behaviour between two linguistic groups predicted by the difference in FTR, over and above the differences in phylogeny and geography?	Yes	Yes	Yes	No	Yes
Partial Stratified Mantel test	As above, but only comparing samples within language families.	Yes	Yes	Yes	No	No
Geographic autocorrelation	Does the relationship between FTR and savings exhibit geographical clustering?	Yes	No	Yes	No	Yes
Phylogenetic Generalised Least Squares	Does FTR predict saving behaviour when controlling for phylogeny?	Yes	Yes	No	No	Yes
PGLS within families	As above, but separately for each language family.	Yes	Yes	No	No	No

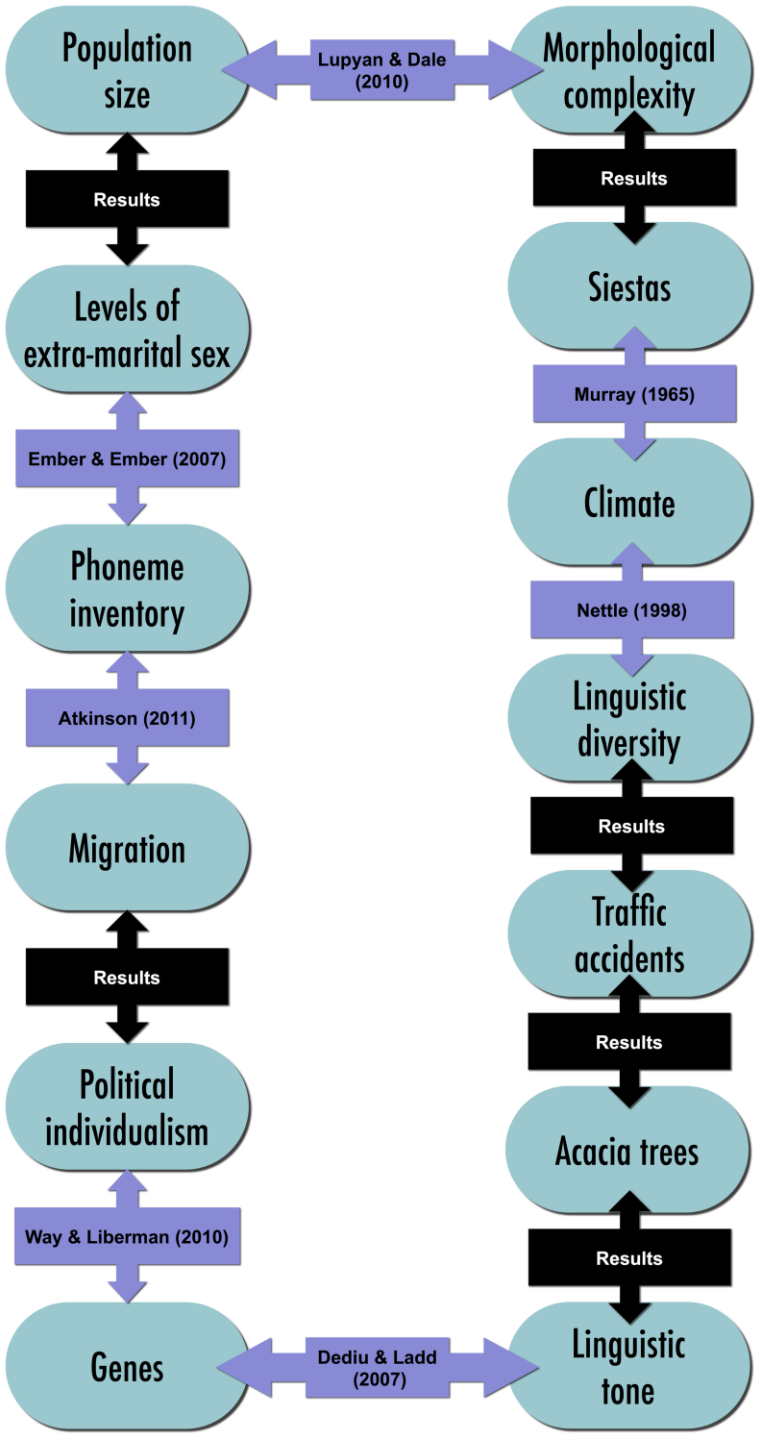
A summary of the statistical methods used to assess whether the relationship between obligatory future tense (FTR) and the propensity to save money is robust to controlling for shared cultural history. Some methods aggregate the data over languages (column 3). Columns 4, 5 and 6 state whether the method implements a control for language family, geographic area and country, respectively. The mixed effects model is the only method that does not aggregate the data and which provides an explicit control for language family, geographic area and country. The final column suggests whether the overall result for the given method demonstrates that the relationship between FTR and savings behaviour is robust. However, this does indicate the status of tests for a given method (see text for details).

doi:10.1371/journal.pone.0132145.t003

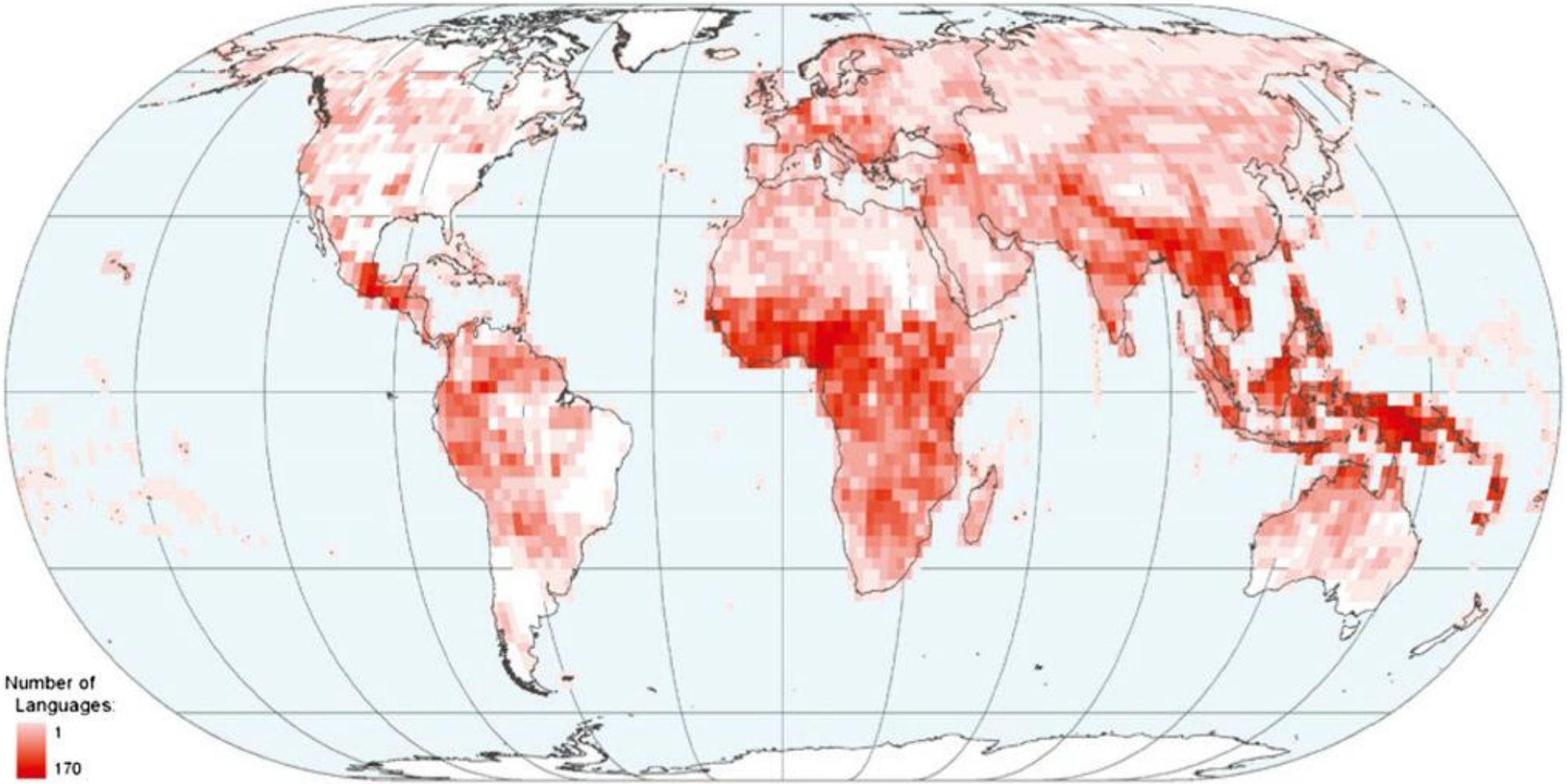
Roberts et al. 2015 >>

False correlations





Lots of problems compounded: Language and the Environment



Gorenflo et al. 2012 >>

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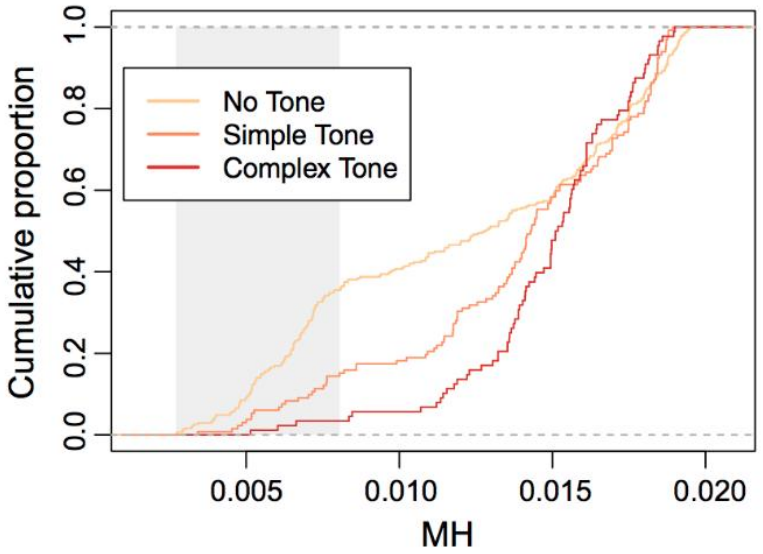
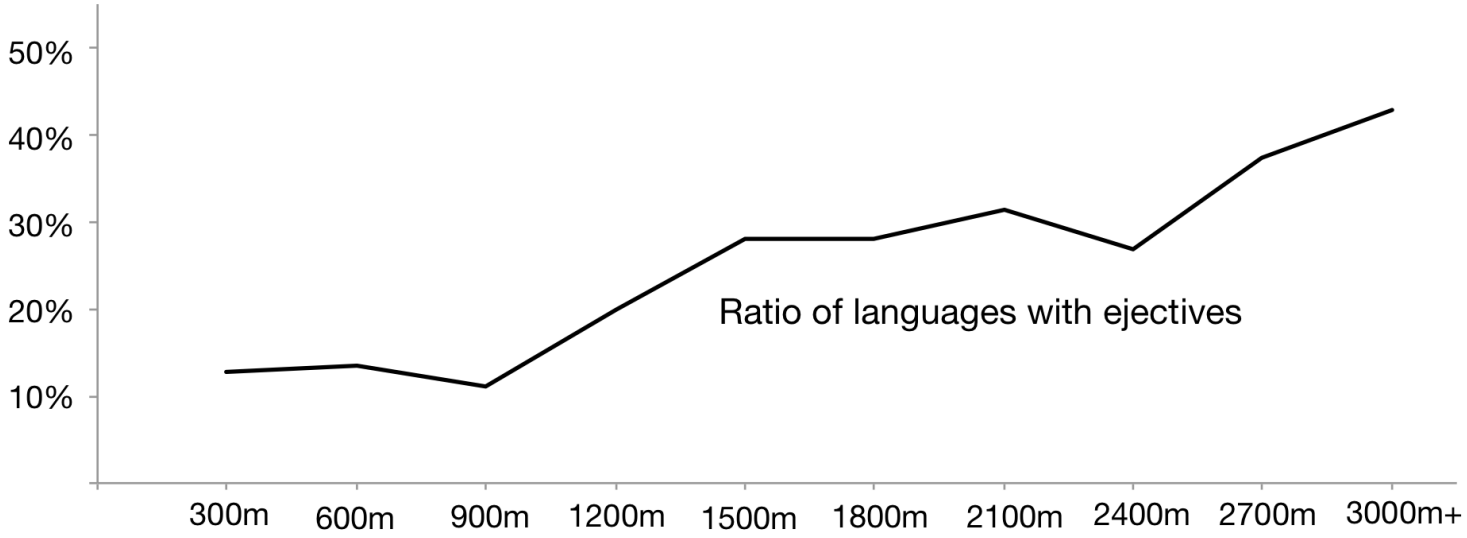
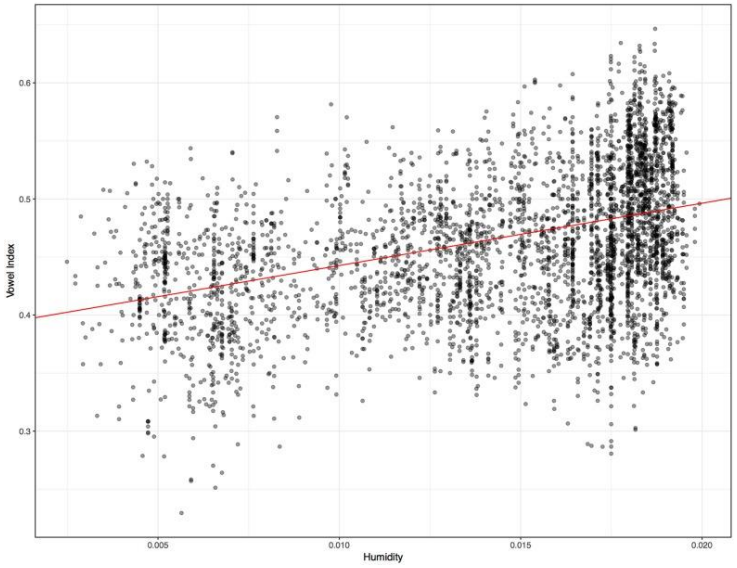






More humid environments have more tonal languages (Everett 2015)
More humid environments have more vowels (Everett 2017)

Higher altitudes have more languages with ejectives (Everett 2013)

Lots of problems compounded: Language and the Environment

[los liβros]

Mexico City
6~27 °C

[loð̞ liβ̞ros]

Tres Ríos
17~27 °C

[lɔ liβ̞rɔ ~ loh liβ̞roh]

San Juan, Puerto Rico
22~31 °C

Lots of problems compounded: Language and the Environment

Table 1
Sonority Scale and Percentages of Sound Classes in 60 Languages

			Percentages of sounds by climate type		
Sound class	Description	Sonority value	Warm	Cold	*t* value[a]
Vowel					
A	low vowels	100	18.0	14.1	2.94***
O	mid back vowels	80	6.6	5.3	1.24
E	mid nonback vowels	69	7.3	7.3	0.00
U	high back vowels	65	7.1	4.7	2.47**
I	high nonback vowels	41	8.7	12.1	–3.38****
Semivowel					
W	high back semivowels	43	1.7	2.1	–0.70
Y	high front semivowels	27	1.6	1.3	0.48
Sonorant consonant					
R	rhotic consonants	36	3.7	2.1	2.02**
L	lateral sonorants	17	3.9	3.1	1.00
N	nasal sonorants	9	11.9	9.8	1.71*
Obstruent					
S	voiceless fricatives	4	5.9	11.1	–4.46****
Z	voiced fricatives	3	1.6	2.3	–1.08
T	voiceless stops	2	13.9	18.5	–2.82***
D	voiced stops	2	8.1	5.8	1.97**
Total percentages			100.0	99.6[b]	

Note: Examples of specific phonetic sounds within each class can be found in Table 1 of Fought et al. (2004).
a. Degrees of freedom = 58 for all comparisons; warm-climate cases, n = 45; cold-climate cases, n =15. All significance tests are two-tailed.
b. Total less than 100 due to rounding.
*$p < .10$. ** $p \leq .05$. ***$p < .01$. ****$p \leq .001$.

Munroe et al. (2009)

Lots of problems compounded: Language and the Environment

(1) Use strong statistical methods like LMERs. Compare effectively between models.

(2) Pay a lot of attention to what goes into our corpora, and consider this as we interpret ours results.

(3) Try to formulate hypothesis a prori and not a posteriori.

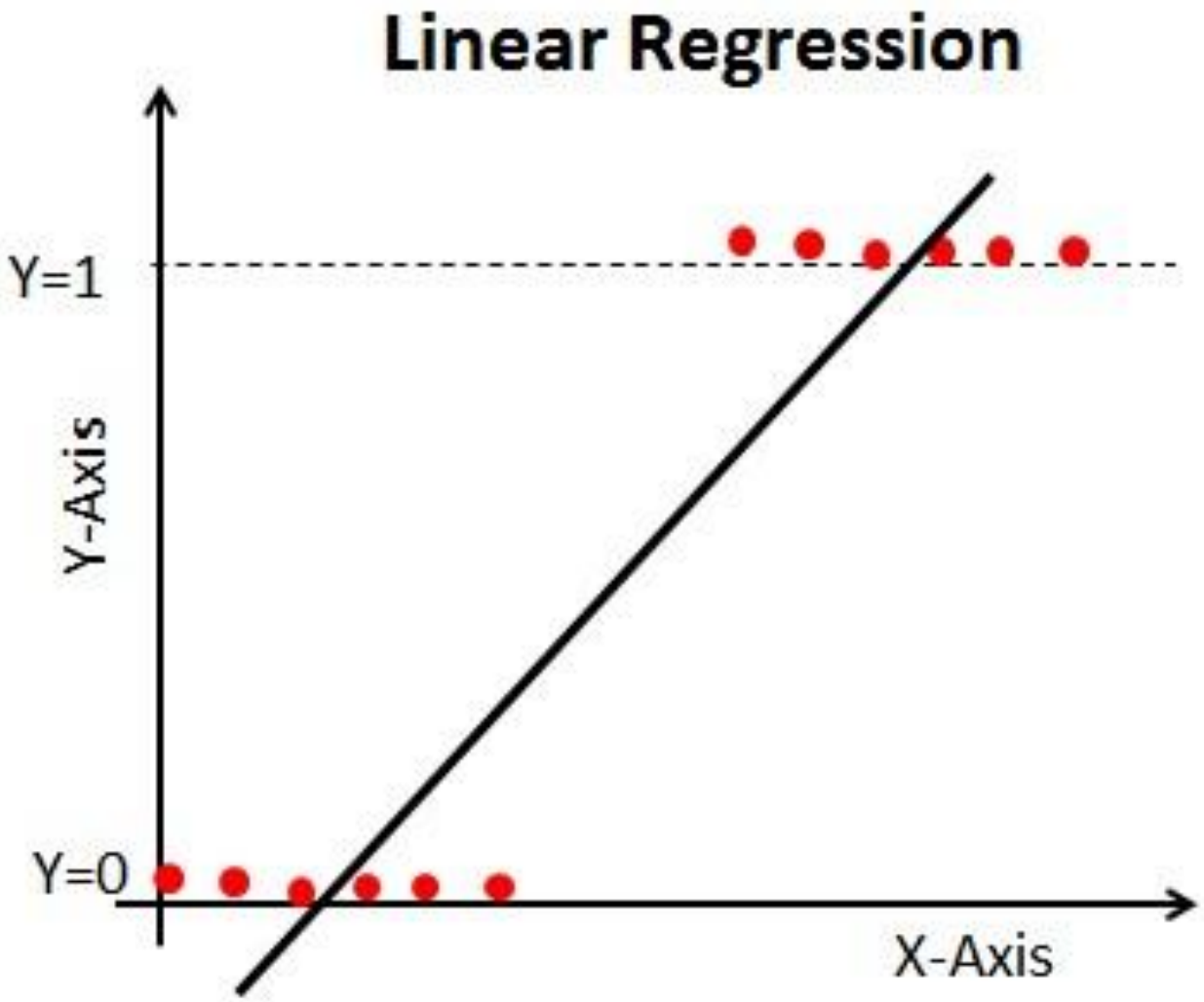
(4) Apply critical thinking to the results that we read about.

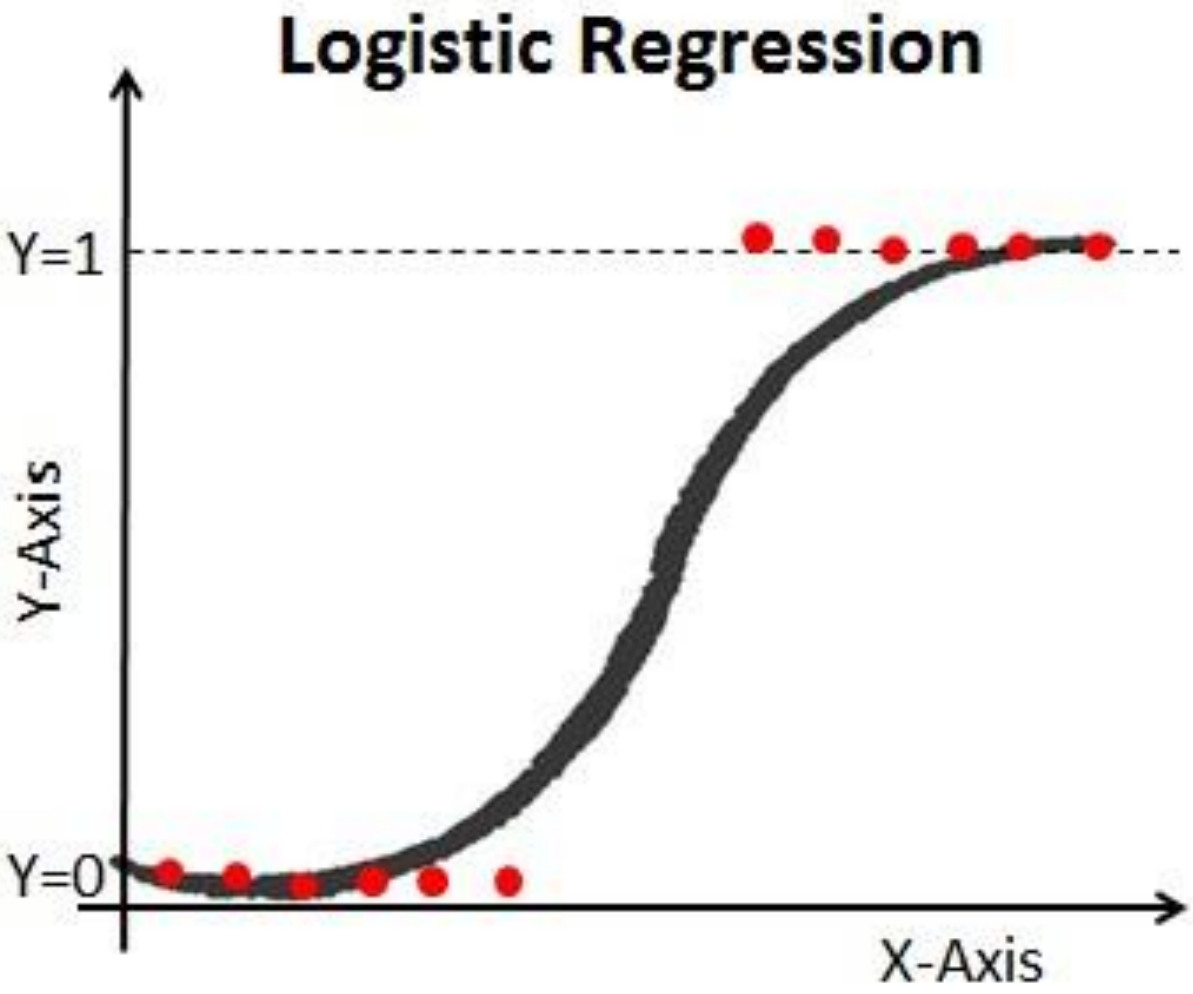
Exercises:

(1) Is speaking a tonal language correlated to the forest coverage of a country? (Exercise 5)
(2) Does speaking tonal languages lead to more freedom of the press? (Exercise 6)

Logistic Regression

Rolando Coto-Solano. LING10
Dartmouth College. February 2026


Linear Regression
Y=1
Y=0
Y-Axis
X-Axis


Logistic Regression
Y=1
Y=0
Y-Axis
X-Axis

Linear Models and Generalized Linear Models (GLMs)

Linear y variable	Count y variable	Categorical y-var with 2 levels	Categorical y-var with 3+ levels
e.g.: Reaction time	e.g.: Visits to the hospital Failures of a microchip Lexical richness between authors	e.g.: Will a patient survive a disease? Will someone take vacations or not? Will speakers of Spanish use /-s/ or not	e.g. How will the patient emerge from an accident? (No awareness, minimal awareness, full awareness). Will someone use *say*, *be like* or *go* for the verb "say"?
Linear model	**GLM: Poisson regression**	**GLM: Binomial (logistic) regression**	**GLM: Multinomial regression**

Logistic Regression: A First Approach

In North American English, the sounds [t] and [d] can sometimes be optionally pronounced as [ɾ] (a “tap”) if followed by a vowel:

For those of you who’d like to ea[t] early, lunch will be served.
For those of you who’d like to ea[ɾ] early, lunch will be served.

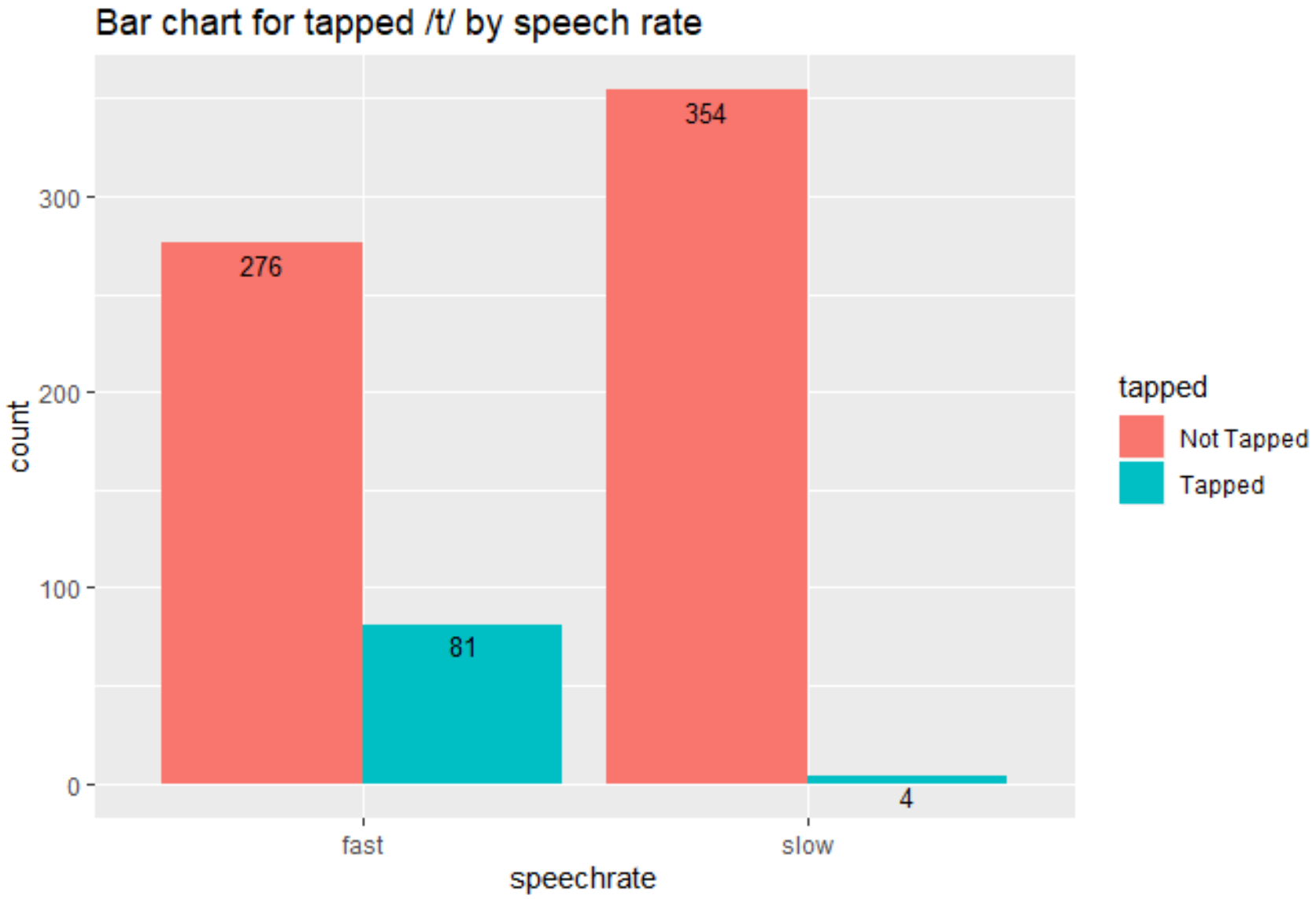
23 students from McGill University Participants were asked to produce sentences like these:

If you plit Alice, John will be mad.

The sentences non-words like *plit* ending in [t] or [d], followed by a vowel initial word (such as “Alice”). The speakers were asked to speak at two different rates: slow versus fast.

How often do participants tap depending on `speakingRate`?

Logistic Regression: A First Approach



Logistic Regression: A First Approach

```
> levels(tap$tapped)
[1] "Not Tapped" "Tapped"

> levels(tap$speechrate)
[1] "fast" "slow"
```

Logistic Regression: A First Approach

```
> mTap = glm(tapped ~ speechrate, data=tap, family=binomial)
> summary(mTap)

Call:
glm(formula = tapped ~ speechrate, family = binomial, data = tap)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.7174  -0.7174  -0.1499  -0.1499   2.9981

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -1.2260     0.1264  -9.701  < 2e-16 ***
speechrateslow -3.2571     0.5185  -6.282 3.34e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of Fisher Scoring iterations: 7
```

Logistic Regression: A First Approach

```
> newdata = rbind( c("fast"), c("slow") )
> newdata = as.data.frame(newdata)
> colnames(newdata) = c("speechrate")
> newdata$speechrate = as.factor(newdata$speechrate)
>
> probsTapped = predict(mTap, newdata, type="response", SE.fit="TRUE")

> probsTapped
         1          2
0.22689076 0.01117318

> probsNonTapped = 1-probsTapped
> probsNonTapped
        1         2
0.7731092 0.9888268
```

Logistic Regression: A First Approach

```
> concordance(mTap)

Call:
concordance.lm(object = mTap)

n= 715
Concordance= 0.7574 se= 0.01515
concordant discordant     tied.x     tied.y    tied.xy
     28674       1104      23772      98028     103677
```

C = 0.5	Practically no association
0.5 < C < 0.7	Little association
0.7 < C < 0.8	Acceptable association
0.8 < C < 0.9	Excellent association
0.9 < C < 1	Outstanding association

Logistic Regression: A First Approach

```
> mTap = glm(tapped ~ speechrate, data=tap, family=binomial)
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -1.2260     0.1264  -9.701  < 2e-16 ***
speechrateslow   -3.2571     0.5185  -6.282 3.34e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How to report this?

The difference between fast and slow speech was studied using a logistic regression over 715 items, with speechrate (levels: fast, slow) as the independent variable, and whether the /t/ is tapped or not as the dependent variable. There was a significant difference between the two speech rates: In slow speech, /t/ was tapped in only 1% of the items, while in fast speech, it was tapped in 23% of the items (β=-3.3 ± 0.5, z=-6.3, p<0.0001, C-index: 0.76).

Logistic Regression: A First Approach through an exercise

The dataset `warlpiri.csv` documents the use of ergative case marking in the narratives of native speakers of Lajamanu Warlpiri describing events in picture books.

We will study two independent variables:

(1) There are 8 children and 13 adults in the set (`AgeGroup`).
(2) Subjects can be animate or inanimate (`AnimacyOfSubject`)

Are there any differences in the use of ergative case that are associated with the age group of the speaker or the animacy of the subject?

Logistic Regression: A First Approach through an exercise

A logistic regression was used, with age group (levels: adults, children) and subject animacy (levels: animate, inanimate) as independent variables and case marking (levels: ergative, other cases) as the dependent variable. The regression included 1055 items. Table 1 shows the percentage of usage of ergative cases. Children use fewer ergatives than adults ($\beta=1.0\pm0.4$, $z=2.7$, $p<0.01$, C-index: 0.63), and animate subjects are more likely to be ergative than inanimate subjects ($\beta=1.4\pm0.5$, $z=2.8$, $p<0.01$). There was no interaction between the two variables ($p=0.09$).

	Animate Subject	Inanimate Subject
Adults	91.5% (140/153)	73.5% (25/34)
Children	80% (104/130)	77% (23/30)

Table 1. Percentage of subjects with ergative marking

Logistic Regression: A Second Approach

```
> mTap = glm(tapped ~ speechrate, data=tap, family=binomial)
> summary(mTap)

Call:
glm(formula = tapped ~ speechrate, family = binomial, data = tap)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.7174  -0.7174  -0.1499  -0.1499   2.9981

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)     -1.2260     0.1264  -9.701  < 2e-16 ***
speechrateslow  -3.2571     0.5185  -6.282 3.34e-10 ***
```

z = β/StError

If z>1.96, it's significant

Logistic Regression: A Second Approach

Linear y variable	Count y variable	Categorical y-var with 2 levels	Categorical y-var with 3+ levels
e.g.: Reaction time	e.g.: Visits to the hospital Failures of a microchip Lexical richness between authors	e.g.: Will a patient survive a disease? Will someone take vacations or not? Will speakers of Spanish use /-s/ or not	e.g. How will the patient emerge from an accident? (No awareness, minimal awareness, full awareness). Will someone use *say*, *be like* or *go* for the verb "say"?
Linear model	**GLM: Poisson regression**	**GLM: Binomial (logistic) regression**	**GLM: Multinomial regression**

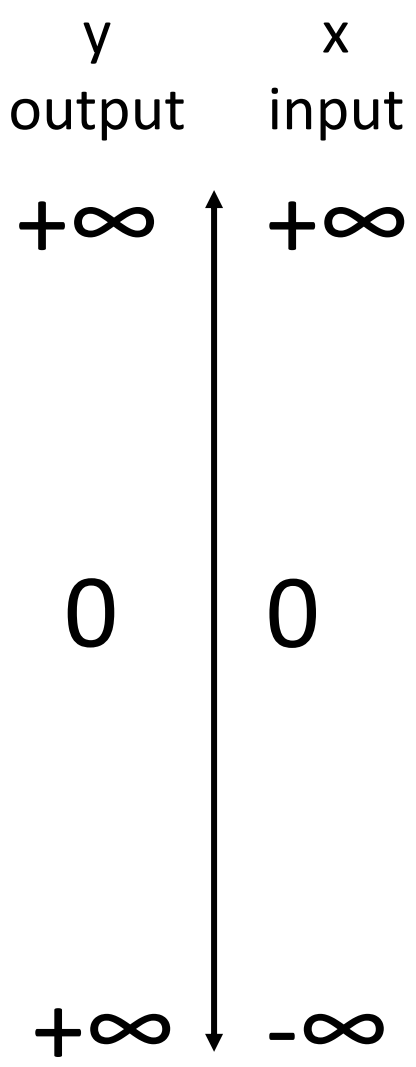
Logistic Regression: A Second Approach

Linear y variable	Count y variable	Categorical y-var with 2 levels	Categorical y-var with 3+ levels
$+\infty$	$+\infty$		
0	1	e.g.: yes/no tapped/not 0/1	e.g. Will someone use *say*, *be like* or *go* for the verb "say"?
$-\infty$	0		
$y_i \in \mathbb{R}$ (y_i can take any value)	$y_i \in \mathbb{N}$ (y_i can only be whole numbers)	Binominal (We want to predict probability)	Multinomial (We want to predict probability)
Linear model	**GLM: Poisson regression**	**GLM: Binomial (logistic) regression**	**GLM: Multinomial regression**

y prediction	linear x predictors
$+\infty$	$+\infty$
0	0
$-\infty$	$-\infty$

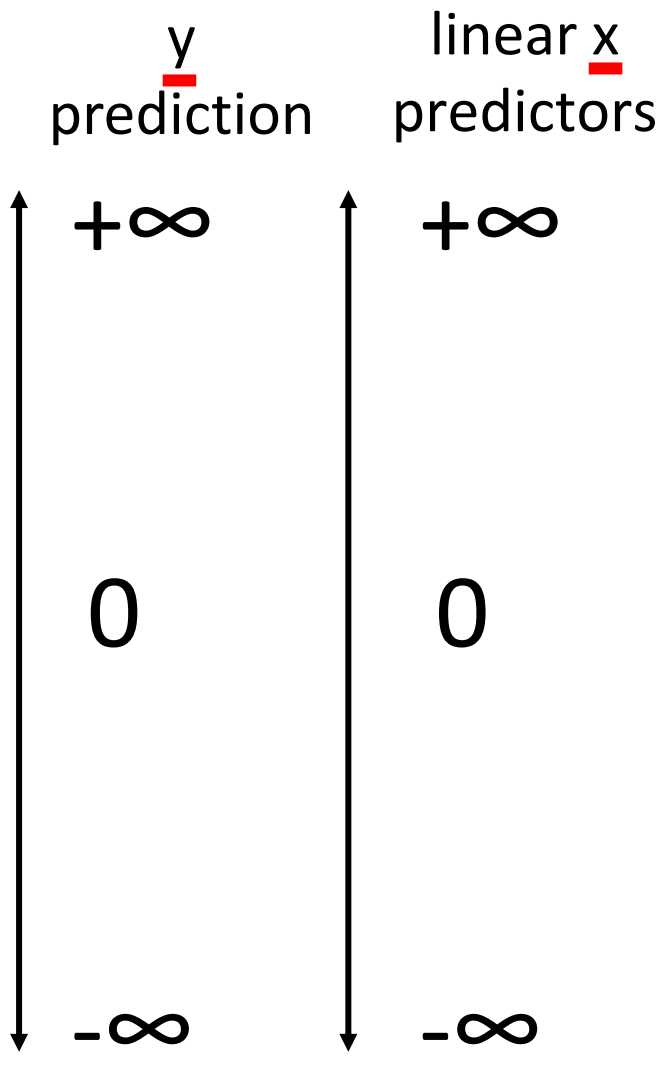
Linear regression:

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon$$



Identity function

$$y = x$$



Linear regression:

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon$$

y output	x input
$+\infty$	$+\infty$
0	0
$+\infty$	$-\infty$

Identity function

$$y = x$$

ln(y)	linear y	linear x predictors
$+\infty$	$+\infty$	$+\infty$
1	0	0
0	$-\infty$	$-\infty$

Natural logarithmic function

$$\ln(y) = x$$

y output	x input
$+\infty$	$+\infty$
0	0
$+\infty$	$-\infty$

Identity function

$$y = x$$

logit(ln(y))	ln(y)	linear y	linear x predictors
1	$+\infty$	$+\infty$	$+\infty$
	1	0	0
0	0	$-\infty$	$-\infty$

Logit function

$$p = \frac{e^{\ln(y)}}{1 + e^{\ln(y)}}$$

If we have a new kind of model such that:

$$\ln(y) = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon$$

This model could calculate numbers from zero to infinity. This would help us make sure that all of our probabilities come out positive.

logit(ln(y))	ln(y)	linear y	linear x predictors
1	$+\infty$	$+\infty$	$+\infty$
	1	0	0
0	0	$-\infty$	$-\infty$

Logit function

$$p = \frac{e^{\ln(y)}}{1 + e^{\ln(y)}}$$

This model has several components:

$$\underbrace{\ln(y)}_{\text{Link function}} = \underbrace{\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon}_{\text{Linear predictor}}$$

In order to find out what our **log odds**, we could also write this equation in the following way:

$$y = e^{(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}$$

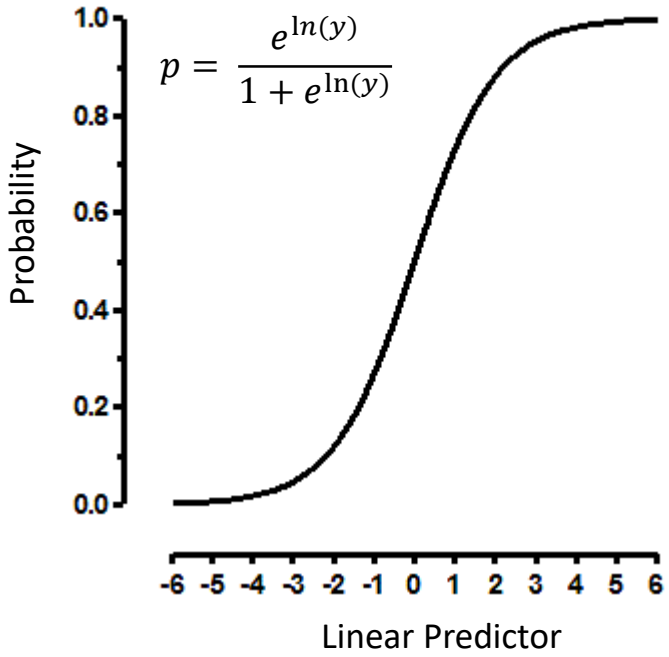
logit(ln(y))	ln(y)	linear y	linear x predictors
1	$+\infty$	$+\infty$	$+\infty$
	1	0	0
0	0	$-\infty$	$-\infty$

Logit function

$$p = \frac{e^{\ln(y)}}{1 + e^{\ln(y)}}$$

Finally, if we run the log odds through the logit function, we can get a probability.

$$p = \frac{e^{\ln(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon)}}{1 + e^{\ln(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon)}}$$



logit(ln(y))	ln(y)	linear y	linear x predictors
1	$+\infty$	$+\infty$	$+\infty$
	1	0	0
0	0	$-\infty$	$-\infty$

Logit function

$$p = \frac{e^{\ln(y)}}{1 + e^{\ln(y)}}$$

So, what is the left side of a logistic regression?

$$\underbrace{\ln(y)}_{\text{Link function}} = \underbrace{\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon}_{\text{Linear predictor}}$$

The result from the linear predictor is just an intermediate step:

In English Tapping:

$\ln(\text{tapped}_{\text{yesTapped}}) =$ $-1.226 - 3.2571(\text{speechrate}_{\text{slow}})$

$\ln(\text{tapped}_{\text{yesTapped/fast}}) =$ $-1.226 - 3.2571*0 =$ $= -1.225952$

$\ln(\text{tapped}_{\text{yesTapped/slow}}) =$ $-1.226 - 3.2571*1 =$ $= -4.483003$

So, what is the left side of a logistic regression?

When we run that result through the logit function, we get the probability:

In English Tapping:

$$tapped_{yesTapped/fast} = \frac{e^{-1.226\ -3.2571*0}}{1 + e^{-1.226-3.2571*0}} = 0.2268$$

$$tapped_{yesTapped/slow} = \frac{e^{-1.226\ -3.2571*1}}{1 + e^{-1.226-3.2571*1}} = 0.001$$

So, what is the left side of a logistic regression?

When we run that result through the inverse of the link, we get the **log odds**:

Link function: Natural logarithm (lm)
Inverse of the link function: Exponential (e^x)

In English Tapping:

$$logOdd\ tapped_{yesTapped/fast} = e^{-1.226\ -3.2571*0} = 0.293$$

$$logOdd\ tapped_{yesTapped/slow} = e^{-1.226\ -3.2571*1} = 0.001$$

So, what is the left side of a logistic regression?

What do the log odds mean? They mean that:

In the FAST speech rate:
For every 1 of the reference level (notTapped), there are 0.293 of the non-reference level (yesTapped).

In the SLOW speech rate:
For every 1 of the reference level (notTapped), there are 0.011 of the non-reference level (yesTapped).

$$logOdd\ tapped_{yesTapped/fast} = e^{-1.226\ -3.2571*0} = 0.293$$

$$logOdd\ tapped_{yesTapped/slow} = e^{-1.226\ -3.2571*1} = 0.011$$

Finally, what are the coefficients?

```
> mTap = glm(tapped ~ speechrate, data=tap, family=binomial)
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -1.2260     0.1264  -9.701  < 2e-16 ***
speechrateslow   -3.2571     0.5185  -6.282 3.34e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The coefficient for speechrateSlow is -3.2571.

exp(-3.2571) = 0.0385

Finally, what are the coefficients?

The coefficient for speechrateSlow is -3.2571.

exp(-3.2571) = 0.0385

The coefficients are the **log-ratio**.

The log-odds of yesTapped/fast are 0.293 (1 notTapped for every 0.293 yesTapped)
The log-odds of yesTapped/slow are 0.011 (1 notTapped for every 0.011 yesTapped)

0.011 / 0.2268 = 0.0385

The exp(beta) is how many times larger the non-reference log-odd is, compared to the reference log-odd.

Logistic Regression: A Second Approach

$$\underbrace{\ln(y)}_{\text{Link function}} = \underbrace{\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon}_{\text{Linear predictor}}$$

Log-odds: $e^{(\text{linearPredictor})}$

For every 1 of the refLevel, there are $e^{(\text{linearPredictor})}$ of the other level.

Logit-function:

$$p = \frac{e^{\ln(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon)}}{1 + e^{\ln(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n + \varepsilon)}}$$

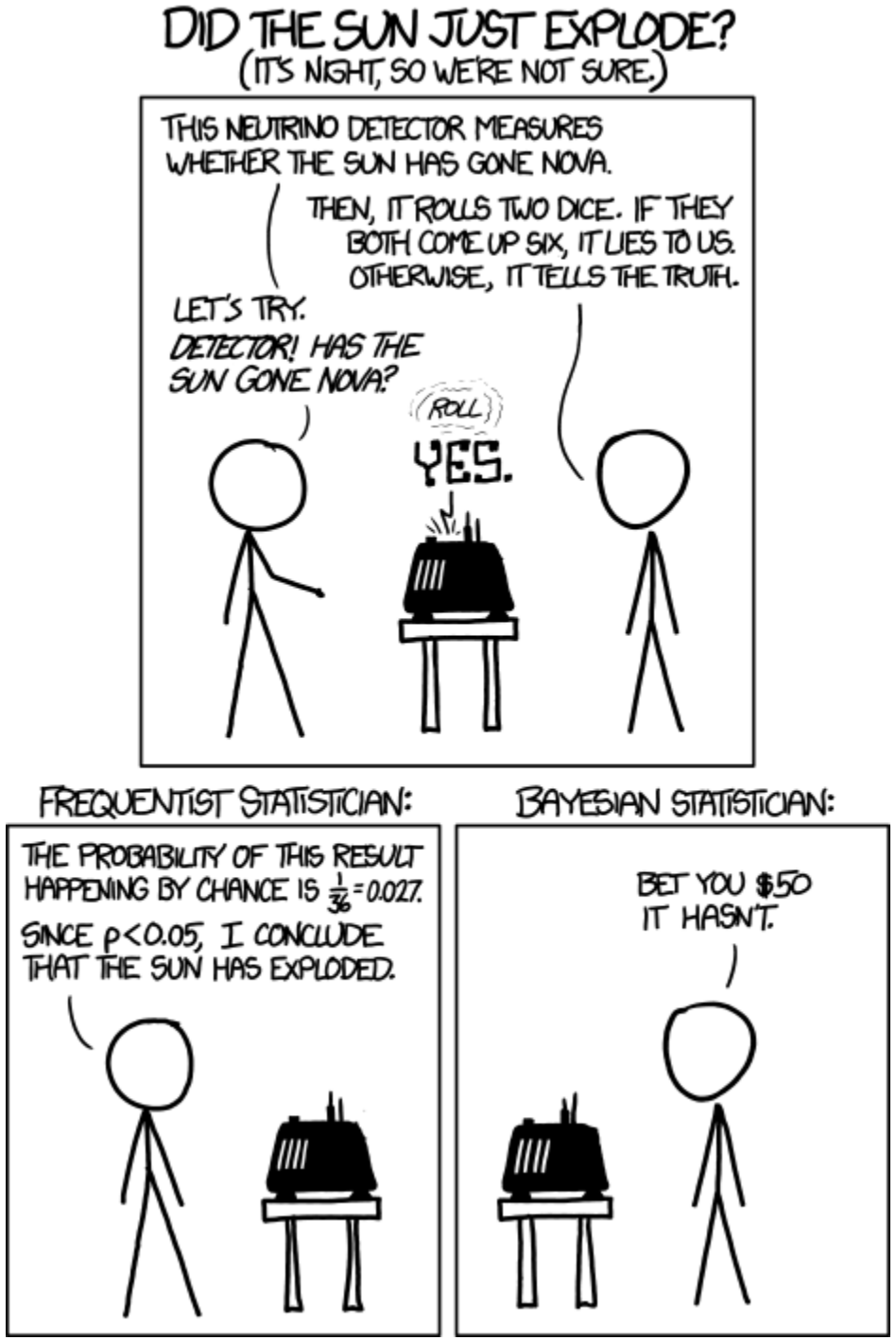
The probability that a certain level occurs in the data.

Log-ratio: $e^{(\beta n)}$

The difference in the ratios between categories in level n and the reference level.

Frequentist and Bayesian Statistics

Rolando Coto Solano. LING10
Dartmouth College. March 2026



https://xkcd.com/1132/

Frequentist Statistics

When we calculate p-values, we try to figure out:

- How many times we see the group X1 (short words with low RT)
- How many times we see the group X2 (long words with low RT)

If $X1_{lowRT}$ is much more frequent than $X2_{lowRT}$
then we believe that there is a significant difference between X1 and X2.

If $X1_{lowRT}$ has the same frequency as $X2_{lowRT}$
then we believe that there is **no** significant difference between X1 and X2.

Frequentist Statistics

When we calculate p-values, we try to figure out:

- How many times we see the group X1 (short words with low RT)
- How many times we see the group X2 (long words with low RT)

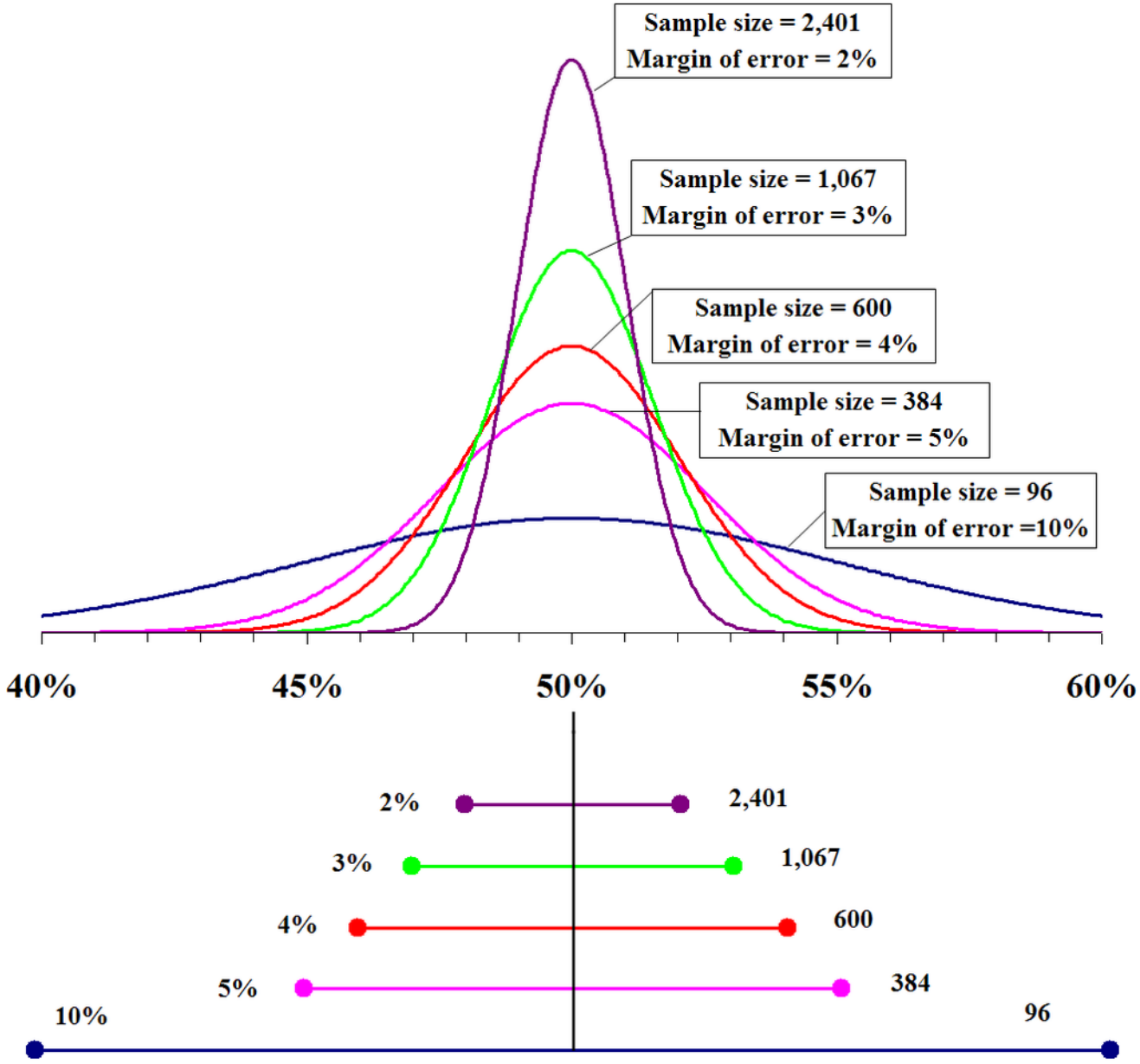
All of our tests (t-tests, ANOVA, chi-square, linear regression, logistic regression) depend on how frequently we see certain values. This family of tests is called **frequentist statistics**.

Problems with Frequentist Statistics

Frequentist Statistics has several problems:

(1) Things can become significant merely by adding more items.

When you add more items, the center of the distribution gets closer to the mean.



Problems with Frequentist Statistics

Frequentist Statistics has several problems:

(1) Things can become significant merely by adding more items.
This also makes the standard error smaller and smaller, and the confidence Interval narrower and narrower!

$$SE = \frac{\sigma}{\sqrt{n}}$$

← Standard deviation

← Number of samples

$$\overline{x} \pm z \frac{s}{\sqrt{n}}$$

Confidence interval

Problems with Frequentist Statistics

Frequentist Statistics has several problems:

(2) Things can become significant merely by adding more items.
As the standard error gets smaller, things will become “more significant”.
(This is also why your normality diagnostics fail when you have many items).

Coefficient in an LM	Standard error	Confidence Interval	Does the confidence Interval include zero?
0.1	0.15	[-0.05 , 0.25]	Yes -> Not significant
0.1	0.05	[0.05 , 0.15]	No -> Significant
0.1	0.005	[0.095 , 0.105]	No -> Really Significant!

Problems with Frequentist Statistics

Frequentist Statistics has several problems:
(2) Being “significantly different” does not necessarily mean that you’re actually different.

Group of people	Pitch (Hz)	Confidence Interval
Group1	136.3	[136.2999, 136.30001]
Group2	136.4	[136.3999, 136.40001]

If I gather 10K samples, these two will be significantly different. But there’s no way this can be linguistically different. Humans couldn’t hear this kind of difference.

Problems with Frequentist Statistics

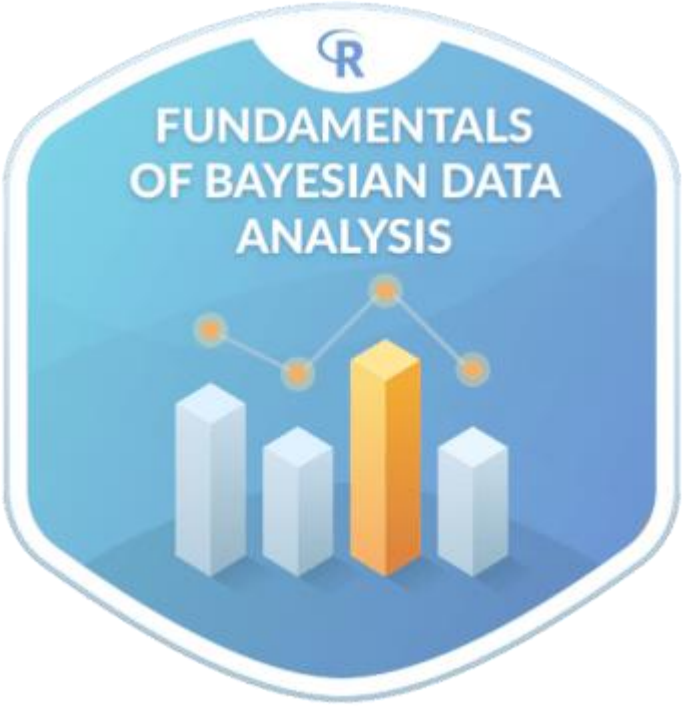
There are numerous problems with frequentist statistics, including:

- When you add items, things will become significant simply because the confidence intervals will become smaller.

- Significant difference does not mean real-world difference.

Problems with Frequentist Statistics

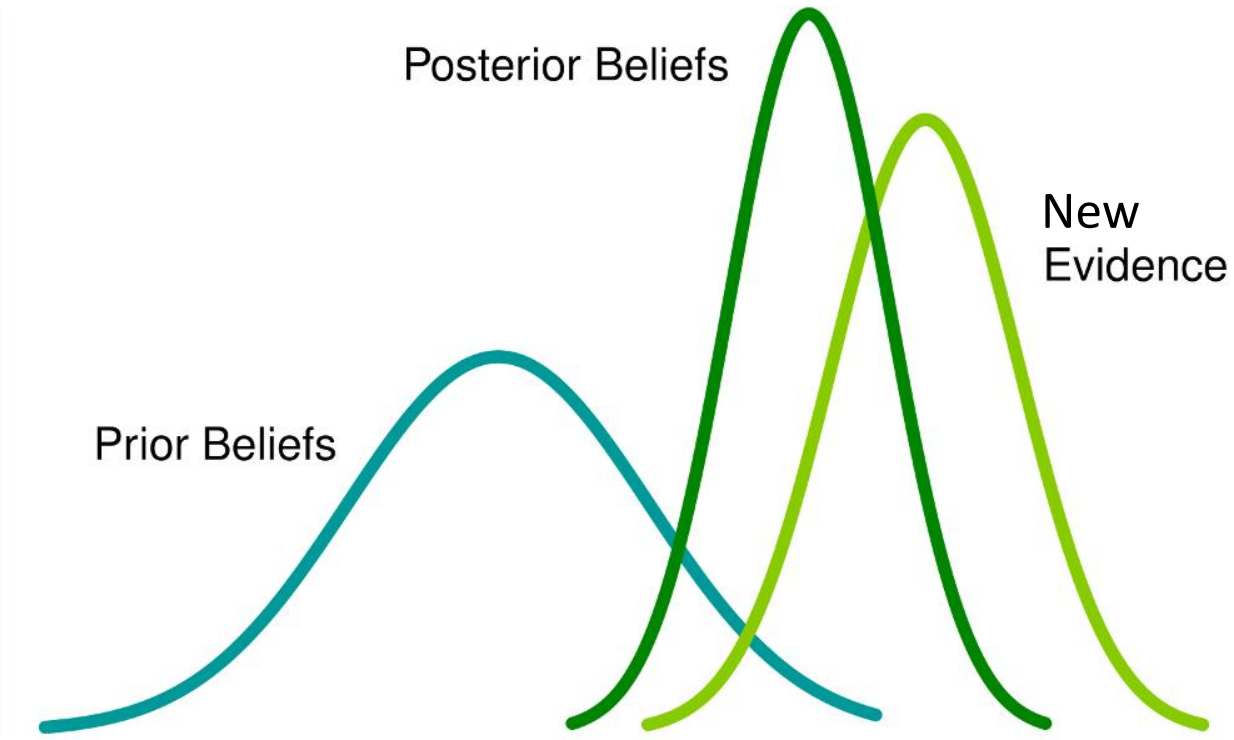
There’s no consensus on what the solution should be.
But many people are trying something called **Bayesian Statistics**.



Is the sun going to rise tomorrow?

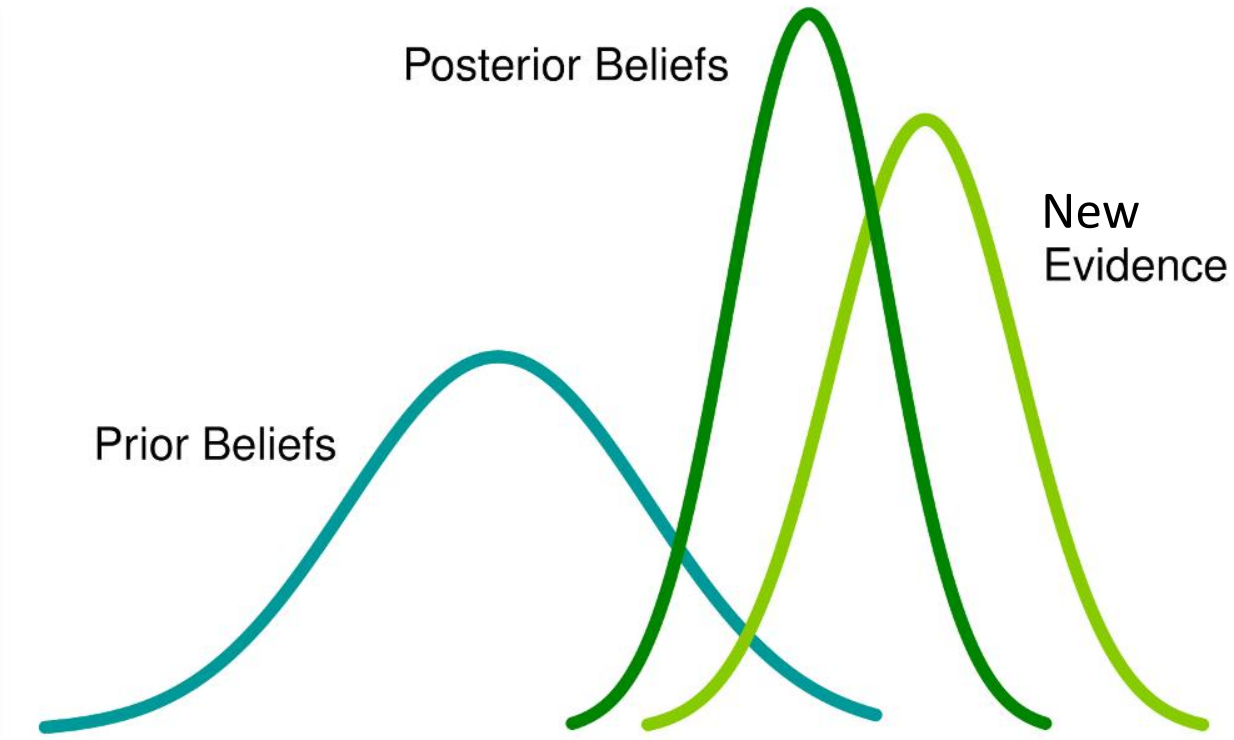
Bayesian Statistics

Our intuition is that the sun **will** rise tomorrow (and that this is not 50%-50%). This is because we are taking **prior information** into account. This is not done easily in the **frequentist** models we have studied.



Bayesian Statistics

When we get new information we update our beliefs. We generate a **posterior**, which describes our beliefs about the situation now.



Bayesian Statistics

“Suppose, out of all the 4 championship races (F1) between Niki Lauda and James Hunt, Niki won 3 times while James managed only 1.

So, if you were to bet on the winner of next race, who would he be ?

Bayesian Statistics

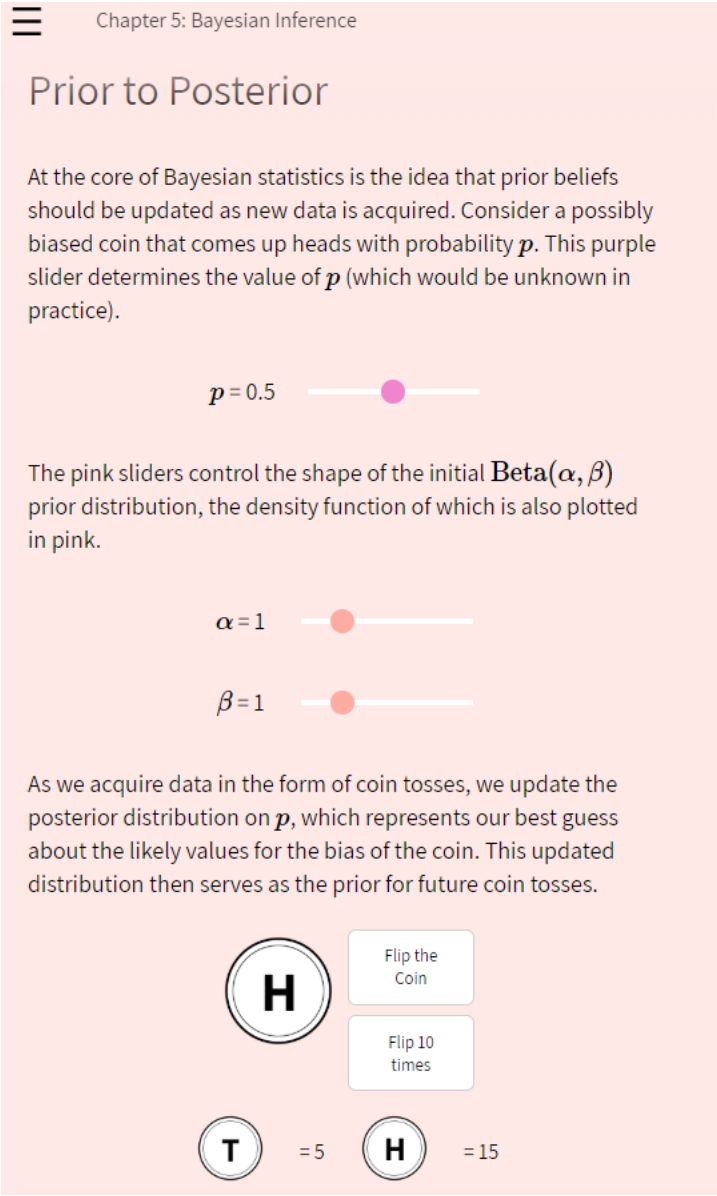
“Suppose, out of all the 4 championship races (F1) between Niki Lauda and James Hunt, Niki won 3 times while James managed only 1.

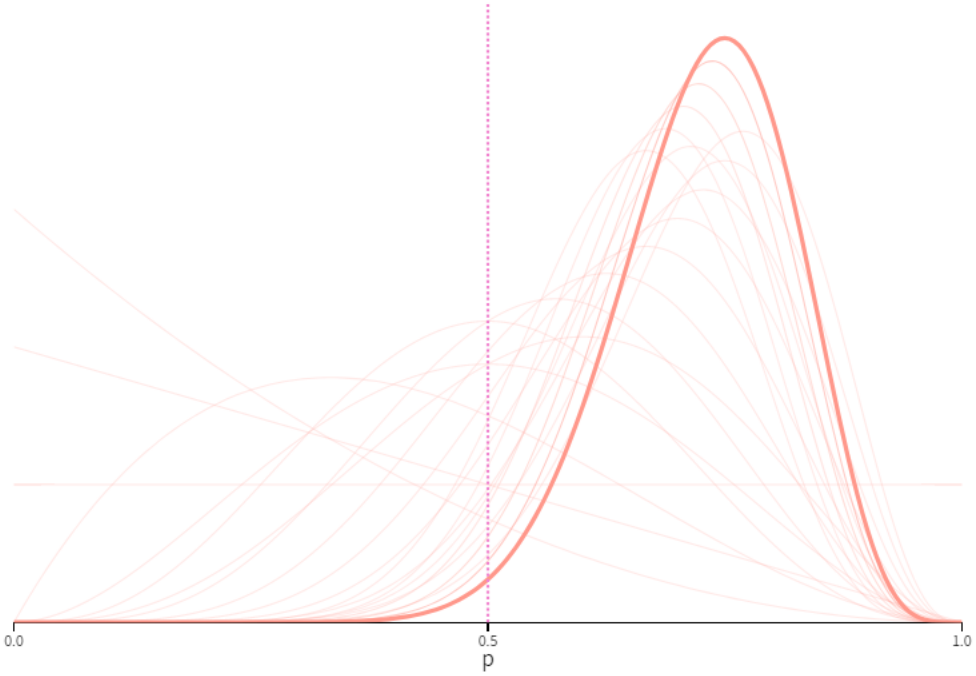
So, if you were to bet on the winner of next race, who would he be ?

I bet you would say Niki Lauda.

Here’s the twist. What if you are told that it rained once when James won and once when Niki won and it is definite that it will rain on the next date. So, who would you bet your money on now?”

Bayesian Statistics

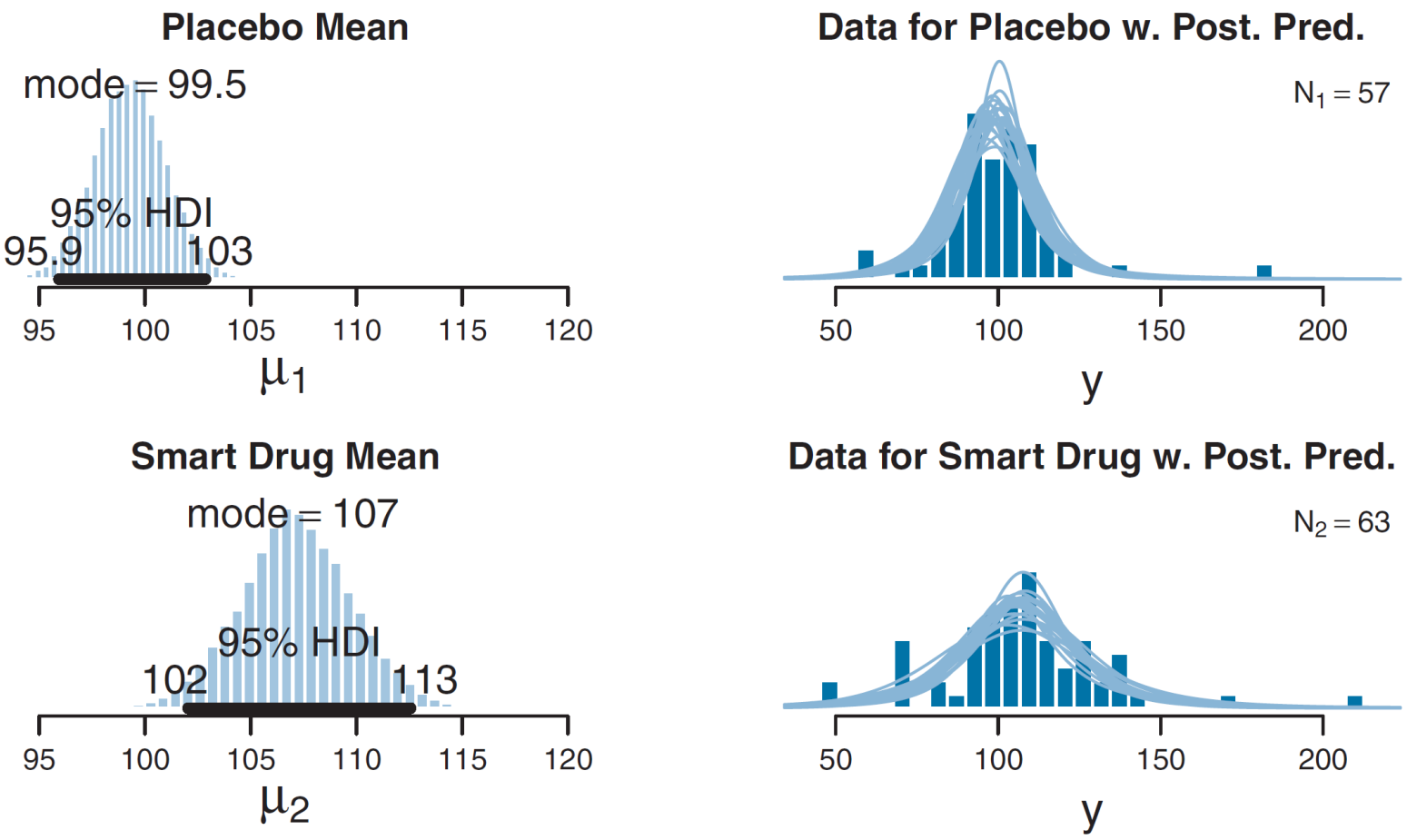




https://seeing-theory.brown.edu/bayesian-inference/index.html#section1

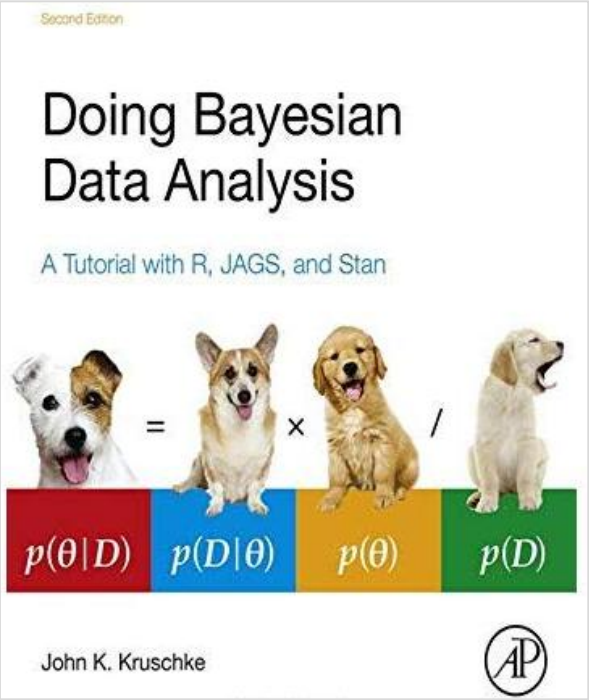
Bayesian Statistics

Our parameters (mean, standard deviation) start from a prior and are then readjusted.

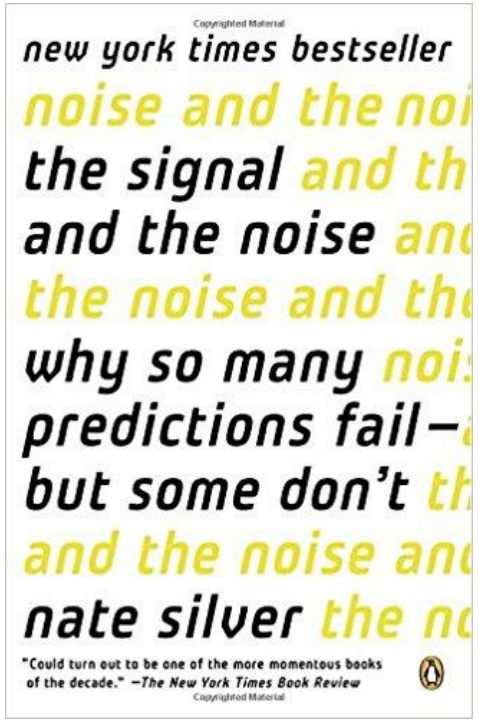


Bayesian Statistics

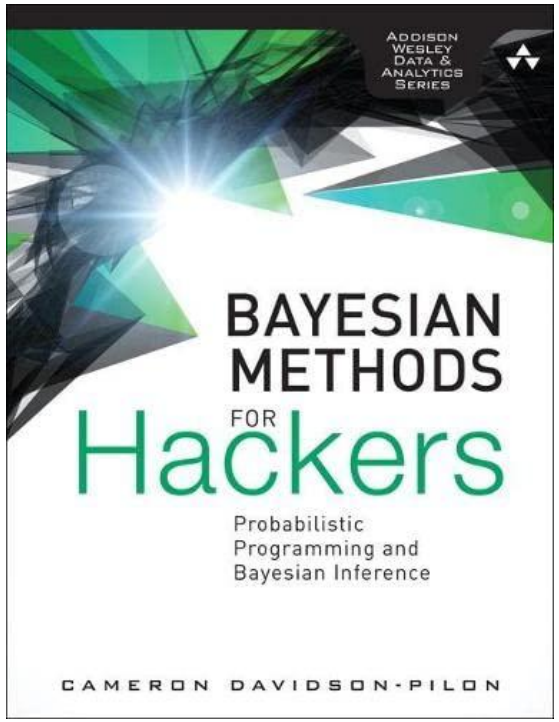
For an academic intro:



For a layperson's intro:



I've heard good things about



https://github.com/CamDavidsonPilon/Probabilistic-Programming-and-Bayesian-Methods-for-Hackers

Bayesian Statistics

https://www.sumsar.net/best_online/

Bayesian Statistics

They both have their advantages and disadvantages:

Frequentist statistics is relatively easy, but its results might be disconnected from reality.

Bayesian statistics takes more of the real world into account, but its mathematics are more complex.

In your career you'll use mostly frequentist statistics, but if you go into a very quantitative field (e.g. psycholinguistics), journals are already starting to ask researchers to include more Bayesian statistics in their papers.

That’s it!

... If you with patient ears attend,
What here shall miss, our toil shall strive to mend.